

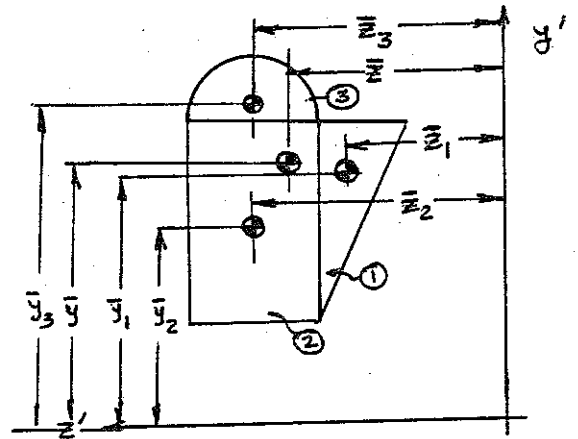
CHAPTER 7

$$\underline{7.1} \quad \bar{y} = \frac{\int y dA}{\int dA} = \frac{\int_1 y dA + \int_2 y dA + \int_3 y dA}{\int_1 dA + \int_2 dA + \int_3 dA}$$

and $\int y dA = \bar{y}_1 A_1$ etc., by definition of $\bar{y}$.

$$\begin{aligned} \text{Hence, } \bar{y} &= \frac{\bar{y}_1 A_1 + \bar{y}_2 A_2 + \bar{y}_3 A_3}{A_1 + A_2 + A_3} \\ &= \frac{\sum \bar{y}_i A_i}{\sum A_i} \end{aligned}$$

Similar proof for $\bar{z}$



7.2

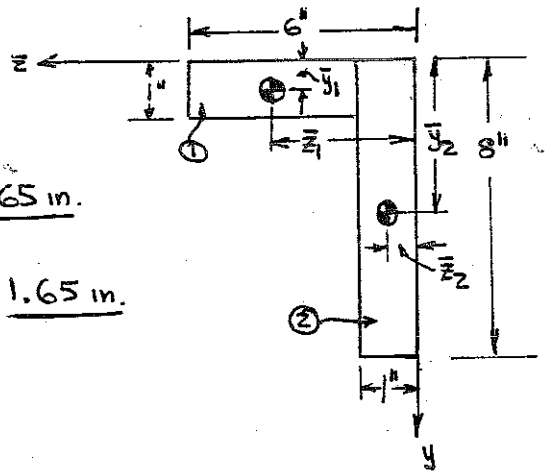
$$A_1 = (6-1)(1) = 5 \text{ m}^2 ; A_2 = (8)(1) = 8 \text{ m}^2$$

$$\bar{y}_1 = 0.5 \text{ m} ; \bar{z}_1 = 3.5 \text{ m}$$

$$\bar{y}_2 = 4 \text{ m} ; \bar{z}_2 = 0.5 \text{ m}.$$

$$\bar{y} = \frac{\sum \bar{y}_i A_i}{\sum A_i} = \frac{(0.5)(5) + (4)(8)}{13} = \underline{2.65 \text{ m}}.$$

$$\bar{z} = \frac{\sum \bar{z}_i A_i}{\sum A_i} = \frac{(3.5)(5) + (0.5)(8)}{13} = \underline{1.65 \text{ m}}.$$



7.3

$$\text{Area} = \int dA = \frac{1}{2}bh.$$

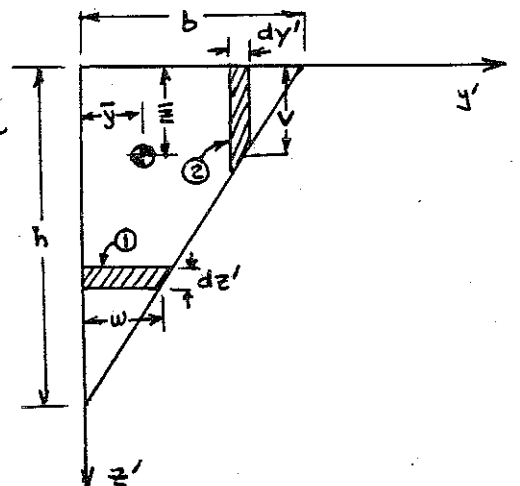
Consider first the elementary area

② :-

$$dA = v dy' \text{ and } v = \frac{b-y'}{b}h$$

$$\therefore \int y' dA = \int_0^b y' \left(\frac{b-y'}{b} \right) h dy' = \frac{hb^2}{6}$$

$$\therefore \bar{y} = \frac{\int y' dA}{\int dA} = \left(\frac{hb^2}{6} \right) \frac{1}{\frac{1}{2}bh} = \frac{b}{3}$$



Similarly, considering the elementary area ①:

$$dA = w dz' \text{ where } w = \frac{h-z'}{h} b$$

$$\int z' dA = \int_0^h z' \left(\frac{h-z'}{h} \right) b dz' = \frac{b h^2}{6}, \text{ and } \bar{z} = \frac{h}{3}$$

We now shift to a new set of coordinate axes, passing through the centroid:

For the area ①:

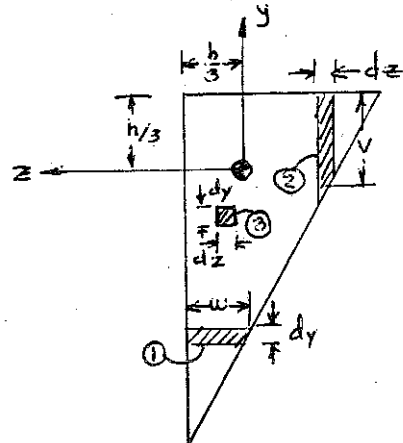
$$dA = w dy \text{ where } w = \frac{2b}{3} + \frac{b}{h} y$$

$$I_{zz} = \int y^2 dA = \int_{-2h/3}^{h/3} y^2 \left(\frac{2b}{3} + \frac{b}{h} y \right) dy = \frac{b h^3}{36}$$

For the area ②:

$$dA = v dz \text{ where } v = \frac{2h}{3} + \frac{h}{b} z$$

$$I_{yy} = \int z^2 dA = \int_{-2h/3}^{b/3} z^2 \left(\frac{2h}{3} + \frac{h}{b} z \right) dz = \frac{b^3 h}{36}$$



To find I_{yz} we will consider the elementary area ③

$$dA = dy dz; \quad I_{yz} = \int yz dA = \int dy \int dz (yz)$$

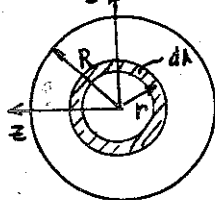
First integrate over z , with y constant, from $z = -(\frac{b}{3} + \frac{b}{h}y)$ to $z = \frac{b}{3}$ and then over y

$$I_{yz} = \int_{-2h/3}^{h/3} dy \int_{-(b/3 + b/h y)}^{b/3} dz (yz) = -\frac{b^2 h^2}{72}$$

7.4

$$I_x = \int r^2 dA = \int (y^2 + z^2) dA$$

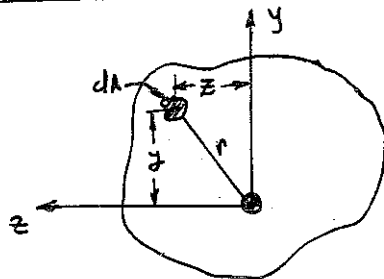
$$= \int y^2 dA + \int z^2 dA = I_{zz} + I_{yy}$$



Symmetry: $I_{yy} = I_{zz}$

and therefore $I_{yy} = I_{zz} = \frac{1}{2} I_x$

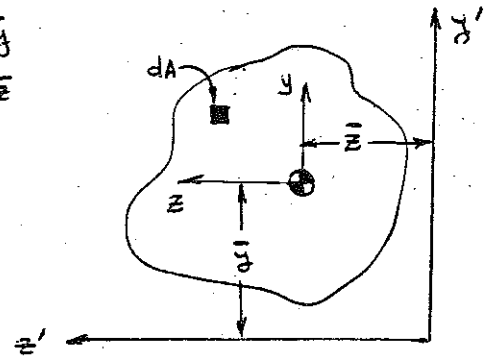
$$I_x = \int r^2 dA = \int_0^R r^2 (2\pi r dr) = \frac{\pi R^4}{2}; \text{ Hence } I_{yy} = I_{zz} = \frac{\pi R^4}{4}$$



7.5

Coordinate Transformation: $\begin{cases} y' = y + \bar{y} \\ z' = z + \bar{z} \end{cases}$

$$\begin{aligned} I_{z'z'} &= \int y'^2 dA = \int (y + \bar{y})^2 dA \\ &= \int y^2 dA + \int 2y\bar{y} dA + \int \bar{y}^2 dA \\ &= I_{zz} + 2\bar{y} \int y dA + \bar{y}^2 A \end{aligned}$$



Now $\int y dA = 0$ since y - z axes are through the centroid

Therefore, $I_{z'z'} = I_{zz} + \bar{y}^2 A$

Similarly, $I_{yy} = I_{yy} + \bar{z}^2 A$

$$I_{y'z'} = \int y'z' dA = \int (y + \bar{y})(z + \bar{z}) dA = \int yz dA + \int \bar{y}z dA + \int \bar{y}\bar{z} dA + \int y\bar{z} dA$$

Again, $\int y dA = \int z dA = 0$

$$\therefore I_{y'z'} = I_{yz} + \bar{y}\bar{z}A$$

7.6

First calculate moments for areas ① and ② about their centroids. Then use parallel axis theorem.

Area ①:

$$I_{zz} = \frac{1}{12}(5)(1)^3 = 0.417 \text{ in}^4$$

$$I_{yy} = \frac{1}{12}(1)(5)^3 = 10.417 \text{ in}^4$$

$$I_{yz} = 0$$

$$A = 5 \text{ in}^2$$

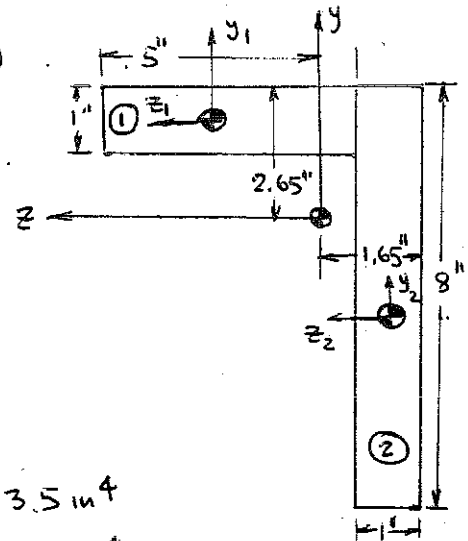
$$\therefore (I_{zz})_1 = 0.417 + (2.15)^2(5) = 23.5 \text{ in}^4$$

$$(I_{yy})_1 = 10.417 + (1.85)^2(5) = 27.5 \text{ in}^4$$

$$(I_{yz})_1 = 0 + (2.15)(1.85)(5) = 19.9 \text{ in}^4$$

Area ②

$$I_{zz} = \frac{1}{12}(1)(8)^3 = 42.67 \text{ in}^4$$



$$I_{y_2 y_2} = \frac{1}{12}(8)(1)^3 = 0.67 \text{ in}^4.$$

$$\bar{I}_{y_2 z_2} = 0.$$

$$A = 8 \text{ m}^2$$

$$\therefore (I_{zz})_2 = 42.67 + (-1.35)^2(8) = 57.3 \text{ in}^4$$

$$(I_{yy})_2 = 0.67 + (-1.15)^2(8) = 11.3 \text{ in}^4$$

$$(I_{yz})_2 = 0 + (-1.35)(-1.15)(8) = 12.4 \text{ in}^4$$

Add results for sections ① and ② :—

$$I_{zz} = 23.5 + 57.3 = 80.8 \text{ in}^4$$

$$I_{yy} = 27.5 + 11.3 = 38.8 \text{ in}^4$$

$$I_{yz} = 19.9 + 12.4 = 32.3 \text{ in}^4$$

7.7 $\bar{y} = \frac{\bar{y}_1 A_1 + \bar{y}_2 A_2}{A} = \frac{-(0)(at) - (a)(at)}{3at} = -\frac{2a}{3}$

(Note: distances of order " λ " are neglected)

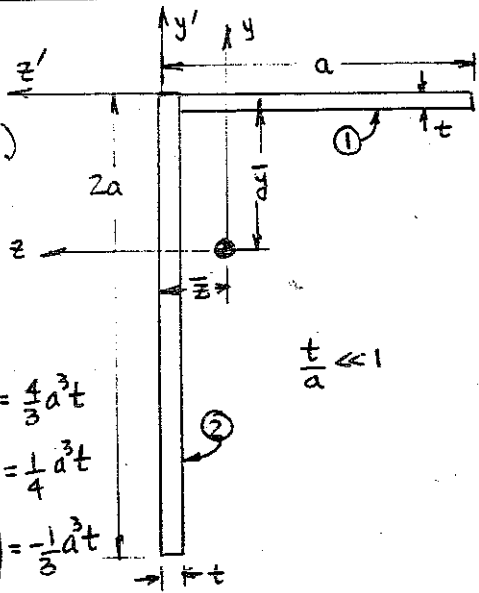
$$\bar{z} = \frac{-(\frac{a}{2})(at) - (0)(2at)}{3at} = -\frac{a}{6}$$

Use parallel axis theorem:

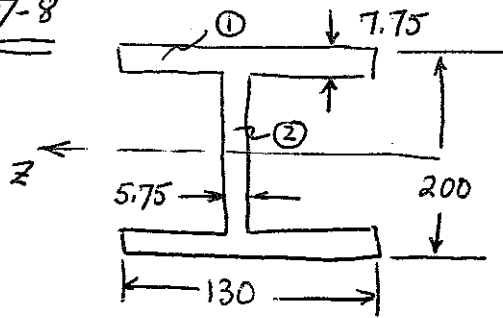
$$I_{zz} = \left[\frac{1}{12} a t^3 + \left(\frac{4a^2}{9} \right) a t \right] + \left[\frac{1}{12} 8a^3 t + \left(\frac{a^2}{9} \right) 2a t \right] = \frac{4}{3} a^3 t$$

$$I_{yy} = \left[\frac{1}{12} t a^3 + \left(\frac{a^2}{9} \right) a t \right] + \left[\frac{1}{12} t^3 2a + \left(\frac{a^2}{9} \right) 2a t \right] = \frac{1}{4} a^3 t$$

$$I_{yz} = \left[0 + \left(\frac{2a}{3} \right) \left(-\frac{a}{3} \right) at \right] + \left[0 + \left(-\frac{a}{3} \right) \left(\frac{a}{6} \right) 2at \right] = -\frac{1}{3} a^3 t$$



7-8



$$I_{zz} = (I_{zz})_2 + 2(I_{zz})_1$$

$$(I_{zz})_2 = \frac{1}{12} 5.75 (184.5)^3 = 3.009 \times 10^6 \text{ mm}^4$$

$$(I_{zz})_1 = \frac{1}{12} (130) (7.75)^3$$

$$+ 130 (7.75) (96.125)^2 = 9.314 \times 10^6$$

$$I_{zz} = 2.164 \times 10^7 \text{ mm}^4$$

7.9 Transformation equations :-

$$n = z \cos \theta - y \sin \theta$$

$$m = z \sin \theta + y \cos \theta$$

$$I_{nn} = \int m^2 dA = \int (z \sin \theta + y \cos \theta)^2 dA$$

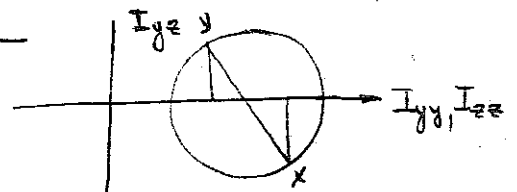
$$= \int z^2 \sin^2 \theta dA + \int y^2 \cos^2 \theta dA + \int 2yz \sin \theta \cos \theta dA$$

$$= I_{yy} \sin^2 \theta + I_{zz} \cos^2 \theta + 2I_{yz} \sin \theta \cos \theta$$

I_{nn} and I_{mm} are obtained in a similar manner.

Using : $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$; $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$; $2\sin \theta \cos \theta = \sin 2\theta$; equations similar to 4.25 are obtained.

Mohr's circle representation :-



7.10 $I_{zz} = 80.8 \text{ in}^4$ Mohr's Circle :-

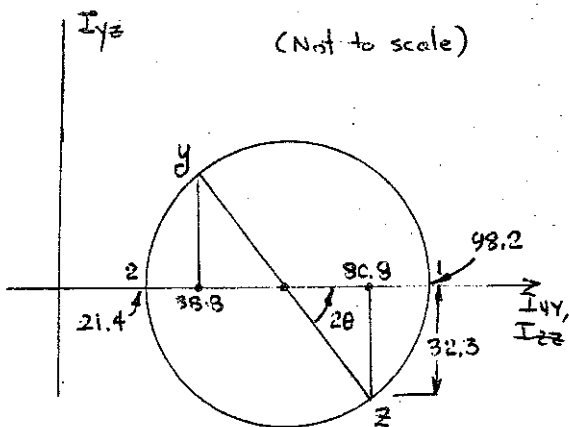
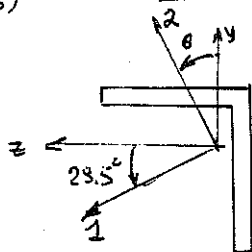
$I_{yy} = 38.8 \text{ in}^4$

$I_{yz} = 32.3 \text{ in}^4$

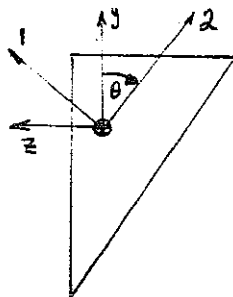
$\tan 2\theta = \frac{32.3}{\frac{1}{2}(80.8 - 38.8)} = 1.54$; $\theta = 28.5^\circ$

$I_{11} = 98.2 \text{ in}^4$

$I_{22} = 21.4 \text{ in}^4$



7.11



From Prob 7.3: Mohr's Circle : (Drawn for $b < h$)

$I_{zz} = \frac{bh^3}{36}$

$I_{yy} = \frac{b^3h}{36}$

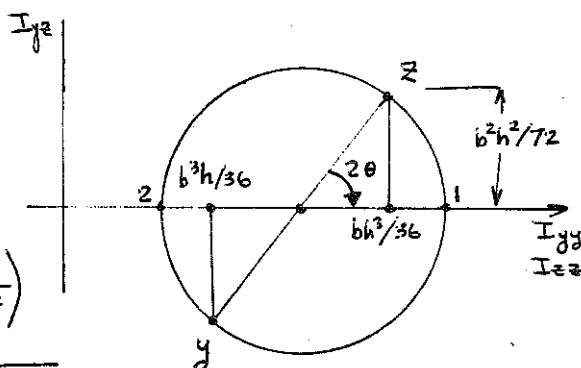
$I_{yz} = -\frac{b^2h^2}{72}$

$\theta = \frac{1}{2} \tan^{-1} \left(\frac{-b^2h^2/72}{\frac{1}{2} \left[\frac{bh^3}{36} - \frac{b^3h}{36} \right]} \right) = \frac{1}{2} \tan^{-1} \left(\frac{-bh}{h^2 - b^2} \right)$

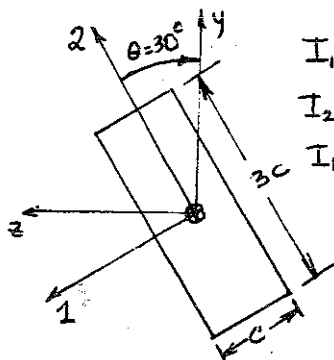
$I_{11} = \frac{1}{2} \left(\frac{bh^3}{36} + \frac{b^3h}{36} \right) + \sqrt{\frac{1}{4} \left(\frac{bh^3}{36} - \frac{b^3h}{36} \right)^2 + \left(\frac{b^2h^2}{72} \right)^2}$

$= \frac{bh}{72} \left[(h^2 + b^2) + \sqrt{(h^2 - b^2)^2 + h^2b^2} \right]$

$I_{22} = \frac{bh}{72} \left[(h^2 + b^2) - \sqrt{(h^2 - b^2)^2 + h^2b^2} \right]$



7.12

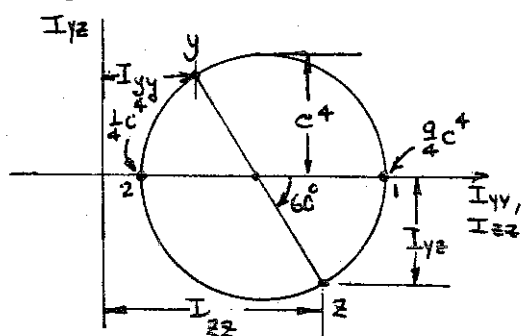


$I_{11} = \frac{1}{12}(c)(3c)^3 = \frac{9}{4}c^4$

$I_{22} = \frac{1}{12}(3c)(c)^3 = \frac{1}{4}c^4$

$I_{12} = 0$

Mohr's Circle :

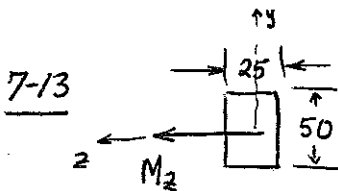


7-12 CONT

$$I_{zz} = \frac{1}{2} \left(\frac{c^4}{4} + \frac{9c^4}{4} \right) + \frac{1}{2} \left(\frac{9c^4}{4} - \frac{c^4}{4} \right) \cos 60^\circ = 1.75 c^4$$

$$I_{yy} = \frac{1}{2} \left(\frac{c^4}{4} + \frac{9c^4}{4} \right) - \frac{1}{2} \left(\frac{9c^4}{4} - \frac{c^4}{4} \right) \cos 60^\circ = 0.75 c^4$$

$$I_{yz} = (c^4) \sin 60^\circ = 0.866 c^4$$

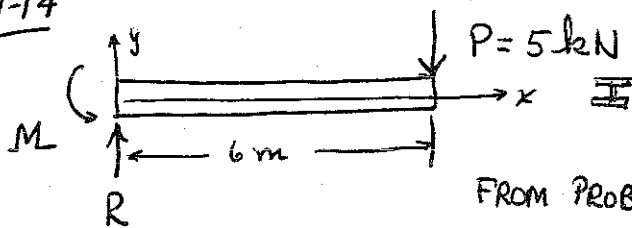


$$M = 1700 \text{ N}\cdot\text{m}$$

$$I_{zz} = \frac{bh^3}{12} = \frac{25(50)^3}{12} = 2.60 \times 10^5 \text{ mm}^4$$

$$\sigma_x = -\frac{M_z y}{I_{zz}} = \frac{-1700 \times 25 \times 10^{-3}}{2.6 \times 10^{-7}} = 163 \text{ MN/m}^2$$

7-14



$$\sum F_y = 0: R = P$$

$$\sum M = 0: M = 5 \times 6 = 30 \times 10^3 \text{ N}\cdot\text{m}$$

$$\text{FROM PROB 7.8 } I_{zz} = 2.164 \times 10^7 \text{ mm}^4$$

$$(\sigma_x)_{\max} = \frac{-M_{\max} y_{\max}}{I_{zz}} = \frac{30 \times 10^3 \times 10^2 \times 10^{-3}}{2.164 \times 10^{-5}} = 138.6 \text{ MN/m}^2$$

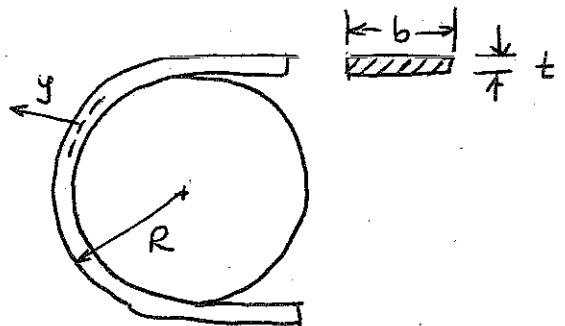
7-15

$$\text{MAX STRAIN} = \frac{y_{\max}}{R} = \frac{t/2}{R}$$

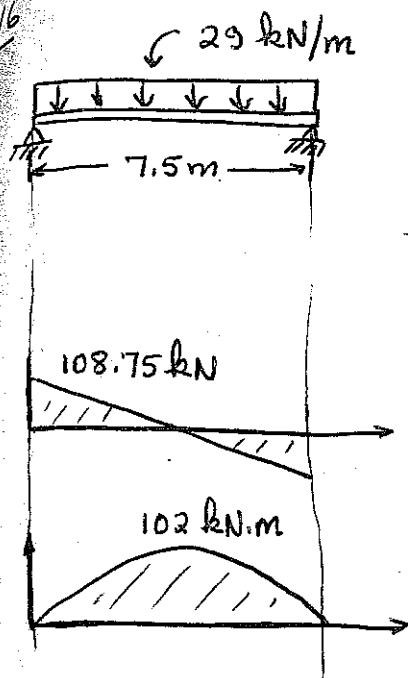
$$\therefore \text{MAX STRESS} = E \frac{y_{\max}}{R} = \frac{Et}{2R}$$

$$t = \frac{2R \sigma_m}{E} = \frac{.3 \times 280}{200 \times 10^3}$$

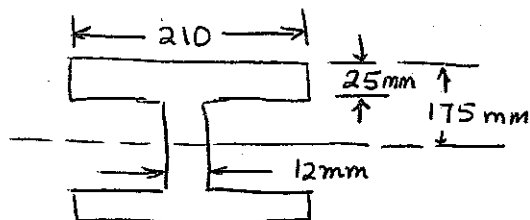
$$= 0.420 \text{ mm}$$



7-7



RESULTANT = 217.5 kN : MAXIMUM MOMENT = 102 kN.m

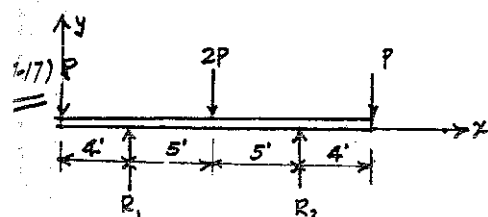


$$I = \frac{bh^3}{12} + 2 \left(\frac{bh^3}{12} + Ad^2 \right)$$

$$= \frac{12(300)^3}{12} + 2 \left(\frac{210(25)^3}{12} + 210 \times 25 \times (162.5)^2 \right)$$

$$= 3.048 \times 10^8 \text{ mm}^4$$

$$\sigma_x = -102 \times 175 \times 10^{-3} / 3.048 \times 10^{-6} = 58.6 \text{ MN/m}^2$$

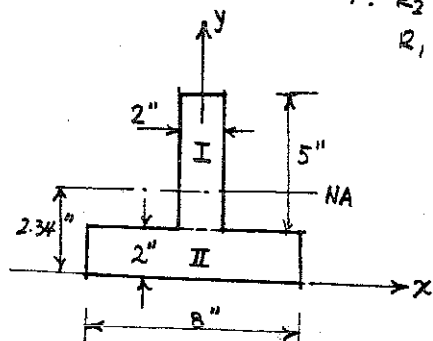
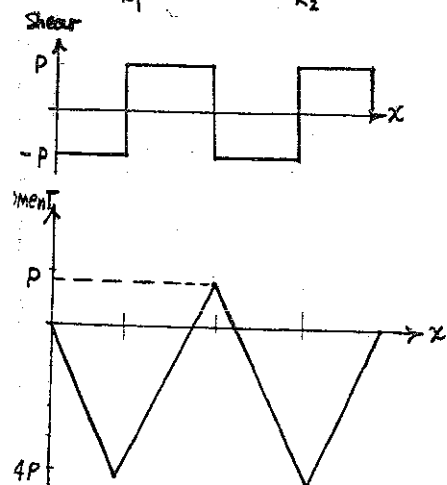


$$\sum F_y = 0 \quad R_1 + R_2 - 4P = 0$$

$$\sum M = 0 \quad 4P \cdot 10 - 10P \cdot 10 + 10R_2 - 14P \cdot 10 = 0$$

$$\therefore R_2 = 2P$$

$$R_1 = 2P$$



$$\bar{Y} \Sigma A = \Sigma AY$$

$$\bar{Y} [(8 \times 2) + (5 \times 2)] = (8 \times 2 \times 1) + (2 \times 5 \times 4.5)$$

$$\bar{Y} = 2.34 \text{ in}$$

$$I_{NA_I} = \frac{bh^3}{12} + Ad^2 = \frac{2 \times 5^3}{12} + (5 \times 2 \times 2.16^2) = 66 \text{ in}^4$$

$$I_{NA_{II}} = \frac{bh^3}{12} + Ad^2 = \frac{8 \times 2^3}{12} + (8 \times 2 \times 1.34^2) = 34.1 \text{ in}^4$$

$$\therefore I_{NA} = I_{xx} = 100.1 \text{ in}^4$$

$$(\sigma_x)_{\max} = -\frac{M_{\max} y_{\max}}{I_{xx}} = 5000 \text{ lb/in}^2 \text{ tension, } 20,000 \text{ lb/in}^2 \text{ compression}$$

$$\therefore P_{\max} = \frac{5000 \times 100.1}{48 \times 4.66} = 2240 \text{ lb.}$$

7.18

$$\bar{y} = \frac{(a)(\frac{3t}{2})(\frac{a}{2}) + (\alpha a)(t)(0)}{(a)(\frac{3t}{2}) + (\alpha a)(t)} = \frac{3a}{4(\alpha + \frac{3}{2})}; t/a \ll 1$$

$$(\sigma_t)_{\max} = \frac{(M_b)_{\max} y_t}{I_{zz}}$$

$$(\sigma_c)_{\max} = \frac{(M_b)_{\max} y_c}{I_{zz}}$$

$$\therefore \frac{(\sigma_t)_{\max}}{(\sigma_c)_{\max}} = \frac{y_t}{y_c} = \frac{1}{3}$$

$$y_t = \bar{y} = \frac{3a}{4(\alpha + \frac{3}{2})}; y_c = a - \bar{y} = \frac{a(4\alpha + 3)}{4(\alpha + \frac{3}{2})}$$

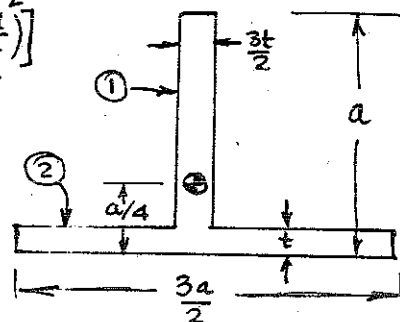
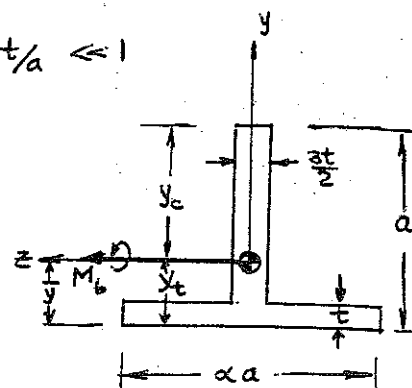
$$\therefore \frac{1}{3} = \frac{3a}{4(\alpha + \frac{3}{2})} \cdot \frac{4(\alpha + \frac{3}{2})}{a(4\alpha + 3)} \text{ giving } \underline{\underline{\alpha = \frac{3}{2}}}$$

For this value of α we have the following section:

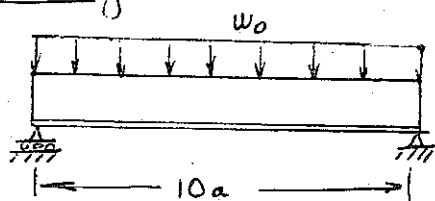
$$I_{zz} = \underbrace{\left[\frac{1}{12} \left(\frac{3t}{2} \right) (a^3) + \left(\frac{3at}{2} \right) \left(\frac{a}{4} \right)^2 \right]}_{\textcircled{1}} + \underbrace{\left[\frac{1}{12} \left(\frac{3a}{2} \right) (t^3) + \left(\frac{3at}{2} \right) \left(\frac{a}{4} \right)^2 \right]}_{\textcircled{2}}$$

$$= \frac{5}{16} a^3 t$$

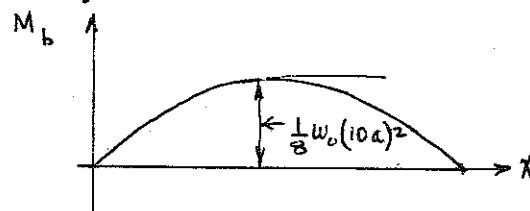
$$y_t = \frac{a}{4}; (\sigma_t)_m = \frac{(M_b)_{\max} \left(\frac{a}{4} \right)}{\frac{5}{16} a^3 t} = \frac{4}{5} \frac{(M_b)_{\max}}{a^2 t}$$



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Bending Moment:



$$(M_b)_{\max} = \frac{25}{2} w_0 a^2$$

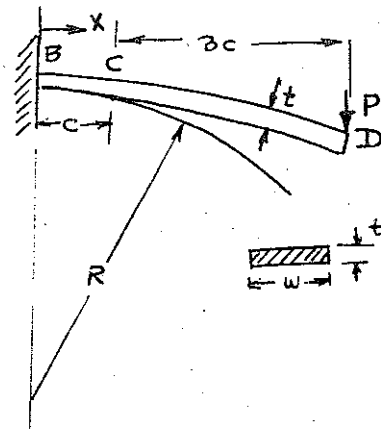
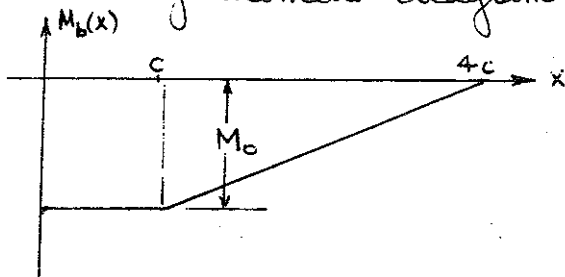
$$\therefore (\sigma_t)_m = \frac{4}{5} \left(\frac{25}{2} w_0 a^2 \right) \frac{1}{a^2 t} = 10 \frac{w_0}{t}$$

$$\text{ie } \underline{\underline{w_0 = \frac{t}{10} \sigma_t}}$$

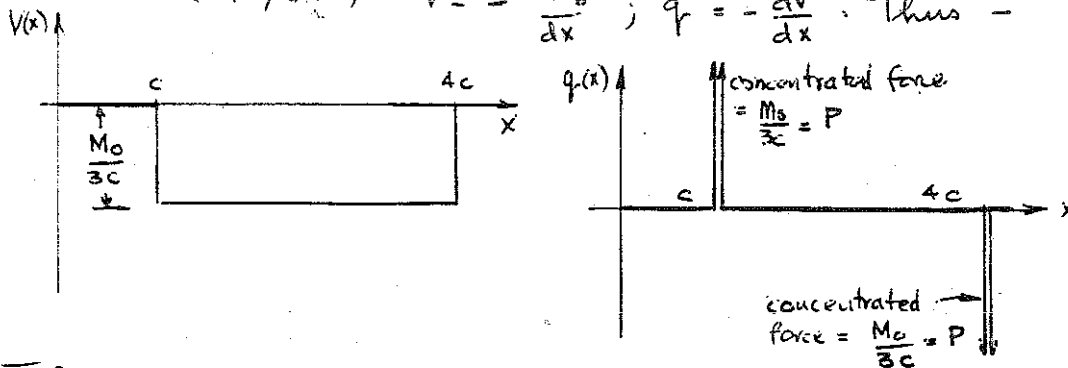
7.19

The strip has constant curvature over the length BC. Therefore the bending moment along BC must be constant and equal to $-M_0 = -\left(\frac{EI}{R}\right) = -\frac{1}{12} \frac{Ewt^3}{R} = -P(3c)$

Bending moment diagram:

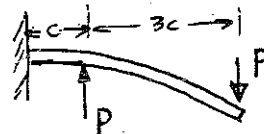


From the bending moment $M_b(x)$ we can successively obtain the shear force $V(x)$ and the loading $q(x)$ by using the relations (3.11, 3.12): $V = -\frac{dM_b}{dx}$; $q = -\frac{dV}{dx}$. Thus -

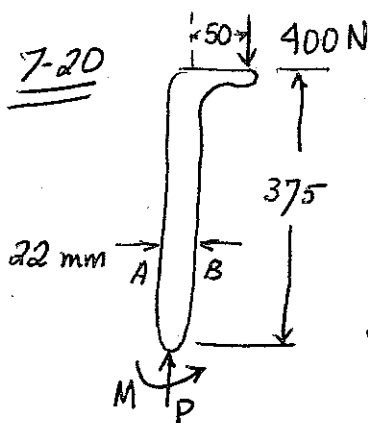


Thus the reaction between strip and the block consists of a concentrated force $= \frac{M_0}{3c}$ at $x=c$.

$$\text{Also, } P = \frac{M_0}{3c} = \left(\frac{1}{3c}\right) \frac{1}{12} \frac{Ewt^3}{R} = \frac{Ewt^3}{36cR}$$



[Note: if (w/t) is large then we must use a plane strain solution (see prob 7.26), which gives $P = \frac{Ewt^3(1-\nu^2)}{36cR}$]



$$A = \pi r^2 = \pi (11)^2 = \pi 121 \text{ mm}^2$$

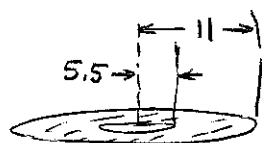
$$I_{zz} = \frac{\pi r^4}{4} = \pi 3660 \text{ mm}^4$$

$$M = 400 \times 50 \times 10^{-3} = 20 \text{ N}\cdot\text{m}$$

$$(a) \quad \sigma_c = \frac{F}{A} = \frac{400}{\pi \times 121 \times 10^{-6}} = 1.05 \text{ MN/m}^2$$

$$\sigma_b = -\frac{My}{I_{zz}} = -\frac{20 \times 11 \times 10^{-3}}{\pi \times 3660 \times 10^{-12}} = 19.13 \text{ MN/m}^2$$

STRESS AT POINT A, $(\sigma_x)_A = 18.08 \text{ MN/m}^2$
 " " " B = 20.18 MN/m²

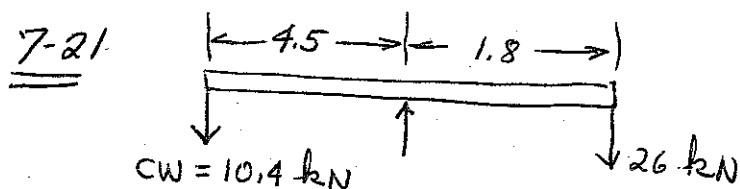


$$(b) \quad A = \pi (11^2 - 5.5^2) = 285 ; I = 1.078 \times 10^{-8} \text{ mm}^4$$

$$\sigma_c = \frac{400 \text{ N}}{285 \times 10^{-6}} = 1.40 \text{ MN/m}^2$$

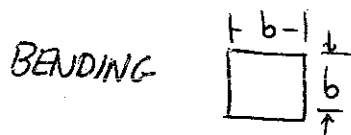
$$\sigma_b = \frac{20 \times 11 \times 10^{-3}}{1.078 \times 10^{-8}} = 20.41 \text{ MN/m}^2$$

(c) INCREASE IN MAXIMUM STRESS $\frac{21.81 - 20.18}{21.81} = 7.47\%$



$$V_{\max} = 26 \text{ kN}$$

$$M_{\max} = 46.8 \text{ kN}\cdot\text{m}$$



$$(\sigma_x)_{\max} = \frac{6 M_{\max}}{b^3}$$

$$b^3 = \frac{6 M_{\max}}{\sigma_{x \max}} = \frac{6 \times 46.8}{52 \times 10^3} = 5.4 \times 10^{-3}$$

$$b = 175 \text{ mm}$$

FOR SHEAR

$$\tau_{\max} = \frac{3}{2} \frac{V_{\max}}{b^2} ; b^2 = \frac{3}{2} \frac{V_{\max}}{\tau_{\max}} = \frac{1.5 \times 26}{10 \times 10^3}$$

$$b = 62.4 \text{ mm}$$

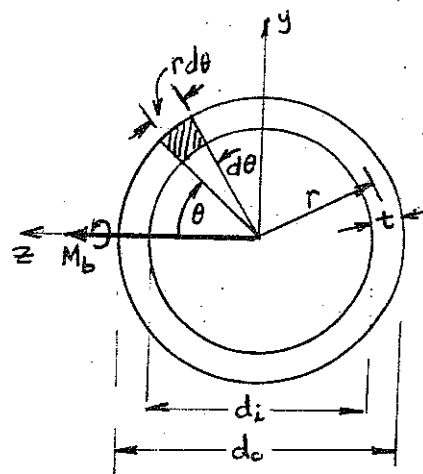
NEED a 175 mm SECTION

7.22

$$\text{Bending stress} = -E y \frac{d\phi}{ds} = -\frac{M_b y}{I_{zz}}$$

$$I_{zz} = \int y^2 dA = 4 \int_0^{\pi/2} (r \sin \theta)^2 (t r d\theta) = \pi r^3 t$$

$$\therefore \sigma_x = -\frac{(M_b)(r \sin \theta)}{\pi r^3 t} = -\frac{M_b \sin \theta}{\pi r^2 t}$$



[Alternate calculation of I_{yy} :-

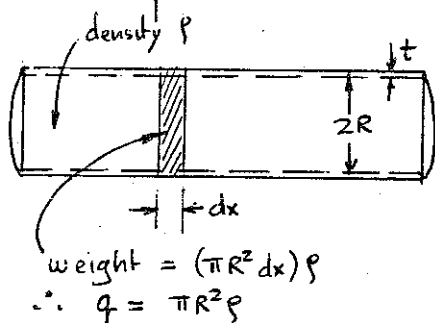
$$I_{yy} = \frac{\pi}{64} d_o^4 - \frac{\pi}{64} d_i^4 = \frac{\pi}{64} [(2r+t)^4 - (2rt)^4]$$

$$= \pi r^3 t + (\text{terms of order } t^2, t^3, t^4)$$

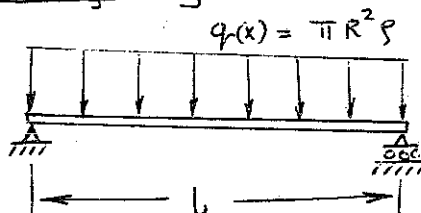
neglect if $\frac{t}{r} \ll 1$

7.23

We will consider the tank as a freely supported beam loaded uniformly along its length. The only stress is that due to bending since internal and external pressures are equal.



Loading diagram:



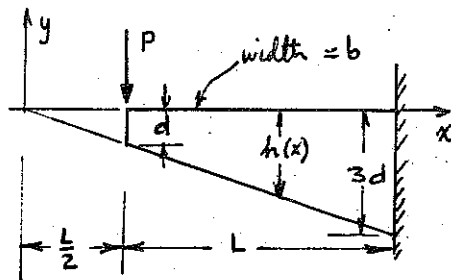
$$(M_b)_{\max} = \frac{1}{4} q L^2 = \frac{\pi R^2 L^2 \rho}{4}$$

From prob. 7.16, max. stress = $\frac{(M_b)_{\max}}{\pi R^2 t} = \frac{\pi R^2 L^2 \rho}{4 \pi R^2 t} = \frac{\rho L^2}{4t}$

This stress is independent of R .

7.24

We will place the maximum stress, at any value of x , equal to $\frac{[M_b(x)][h(x)]}{I_{zz}(x)}$. This results from the assumption that plane cross sections remain plane, (b/d) is small, etc.



$$M_b(x) = P(x - \frac{L}{2}) \quad ; \quad I_{zz}(x) = \frac{1}{12} b h^3 = \frac{1}{12} b \left[2 \frac{x}{L} d \right]^3 = \frac{2}{3} \frac{b x^3 d^3}{L^3}$$

$$\therefore (\sigma_x)_{\max} = \frac{P(x - \frac{L}{2})(\frac{x}{L} d)}{\frac{2}{3} \frac{b x^3 d^3}{L^3}} = \frac{3}{2} \frac{P L^2}{b d^2} \frac{1}{x^2} (x - \frac{L}{2})$$

This will be maximised at the point where:

$$\frac{d}{dx} \left[\frac{1}{x^2} (x - \frac{L}{2}) \right] = 0 \quad \text{and} \quad \frac{d^2}{dx^2} \left[\frac{1}{x^2} (x - \frac{L}{2}) \right] < 0 \quad ; \quad \text{giving } \underline{x = L}$$

$$\text{and, max. stress} = \frac{3}{4} \frac{P L}{b d^2}$$

7.25

We will assume that the shell is stiff enough to permit simple bending, i.e., plane sections remain plane.

$$\text{Then, } \sigma_{\max} = \frac{M_b y_{\max}}{I_{zz}}$$

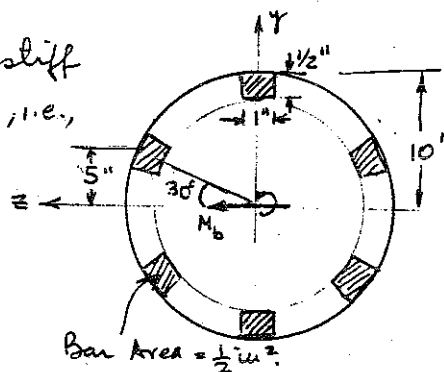
Calculation of I_{zz} :

1) I_{zz} of shell neglected.

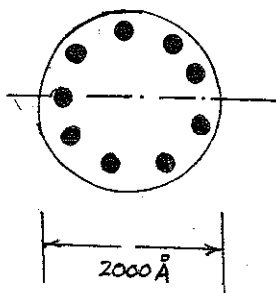
2) I_{zz} of each bar = (I about bar centroid) + $A \bar{y}^2$
negligible.

$$\therefore I_{zz} = \left(\frac{1}{2} \right) [(10)^2 + (10)^2 + (5)^2 + (5)^2 + (5)^2 + (5)^2] = 150 \text{ in}^4$$

$$\therefore \sigma_{\max} = \frac{100,000 (10)}{150} = 6700 \text{ psi}$$



(7-26)



$$M = 5 \times 10^{-7} \text{ N} \cdot \text{m}$$

$$\rho = 6 \times 10^{-6} \text{ m}$$

$$I_{zz} = 4 \times 10^{-8} \text{ mm}^4 = 4 \times 10^{-20} \text{ m}^4$$

$$E = \frac{M \rho}{I_{zz}} = \frac{5 \times 10^{-7} (6 \times 10^{-6})}{4 \times 10^{-20}} = 75 \text{ MN/m}^2$$

(7.27) From arguments similar to those in section 7.6, we have

$$\tau_{zy} = \frac{4P}{3\pi r^4} (r^2 - y^2) \quad \text{note } \int \tau_{zy} dA = P$$

Strain energy
due to
shear

$$U_s = \int \frac{\tau_{zy}^2}{2G} dv$$

$$= \frac{16P^2}{18\pi^2 r^8 G} \int (r^2 - y^2)^2 dv$$

$$= \frac{8P^2}{9\pi^2 r^8 G} \int dz \int_{-r}^r (r^2 - y^2) dy \int_0^{\sqrt{r^2 - y^2}} dx = \frac{4 \cdot 8P^2}{9\pi^2 r^8 G} 2\pi r R \int_0^r (r^2 - y^2)^{3/2} dy$$

let $y = r \sin \phi$

$dy = r \cos \phi d\phi$

$$\int_0^r (r^2 - y^2)^{3/2} dy = \int_0^{\pi/2} r^5 (1 - \sin^2 \phi)^{3/2} r \cos \phi d\phi = \int_0^{\pi/2} r^6 \cos^4 \phi d\phi$$

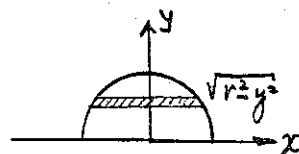
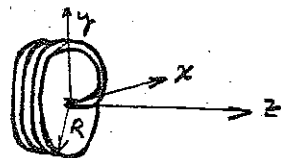
$$= r^6 \frac{3.5\pi}{6.16}$$

$$\therefore U_s = \frac{4 \cdot 8P^2}{9\pi^2 r^8 G} \cdot 2\pi r R \cdot r^6 \frac{3.5\pi}{6.16}$$

$$= \frac{10}{9} \frac{P^2 r R}{r^2 G}$$

$$U_{\text{torsion}} = \frac{P^2 R^3}{2G I_z} 2\pi r R = \frac{P^2 R^3 2\pi}{G r^4} \quad \text{where } I_z = \frac{\pi r^4}{2}$$

$$\frac{U_s}{U_t} = \frac{10}{9} \frac{P^2 r R}{r^2 G} \cdot \frac{G r^4}{2\pi P^2 R^3} = \frac{5}{9} \left(\frac{r}{R}\right)^2$$



7.28 Trial I: $P = \text{about } 1 \text{ lb}$, with perfectly sharp point.

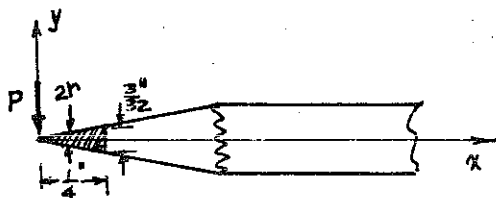
$$\sigma = \frac{My}{I_{zz}}; \quad M = Px$$

$$r = \frac{3}{64} \left(\frac{K}{V_4}\right) = \frac{3}{16} x$$

$$y = r$$

$$I_{zz} = \frac{\pi r^4}{4} = \frac{\pi}{4} \left(\frac{3}{16}\right)^4 x^4$$

$$\sigma(x) = \frac{(Px) \left(\frac{3}{16} x\right)}{\frac{\pi}{4} \left(\frac{3}{16}\right)^4 x^4} = 192 \frac{P}{x^2}$$



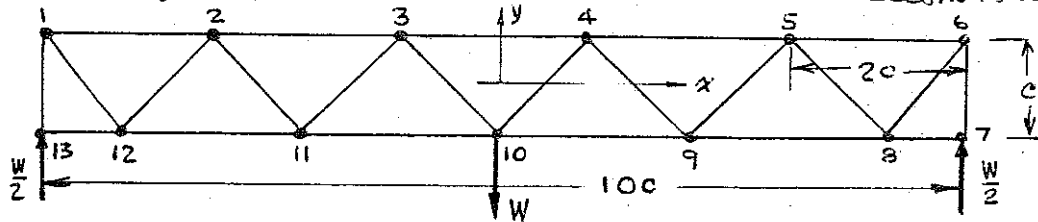
Hence a perfectly sharp point will always tend to break off as $x \rightarrow \infty$ as $x \rightarrow 0$

Trial II: Assume $\frac{1}{64}$ " flat on end due to mechanics of sharpener or breaking off as above.

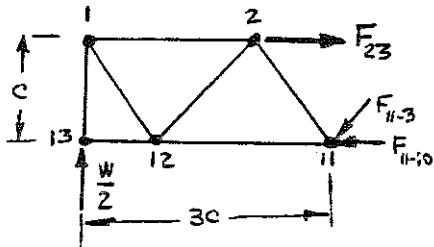
$P = 1 \text{ lb}$, as before.

7.30

Consider the truss to be pin-jointed.
Overall equilibrium leads to the end reactions shown.



Sample free body diagram :-



Equilibrium:

$$\sum M_{11} = 0 \quad (\text{pin joint transfers no moment})$$

$$\therefore \frac{W}{2}(3c) + F_{23}(c) = 0$$

$$F_{23} = -\frac{3W}{2}$$

In a similar manner we obtain :-

$$F_{12} = -\frac{W}{2} ; F_{23} = -\frac{3W}{2} ; F_{34} = -\frac{5W}{2} ; F_{45} = -\frac{3W}{2} ; F_{56} = -\frac{W}{2} ;$$

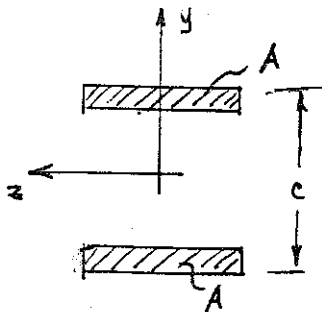
$$F_{78} = 0 ; F_{89} = \frac{2W}{2} ; F_{910} = \frac{4W}{2} ; F_{1011} = \frac{4W}{2} ; F_{1112} = \frac{2W}{2} ; F_{1213} = 0 ;$$

The corresponding stresses are $\sigma = F/A$:-

$$\sigma_{12} = -\frac{W}{2A} ; \sigma_{23} = -\frac{3W}{2A} ; \sigma_{34} = -\frac{5W}{2A} ; \sigma_{45} = -\frac{3W}{2A} ; \sigma_{56} = -\frac{W}{2A}$$

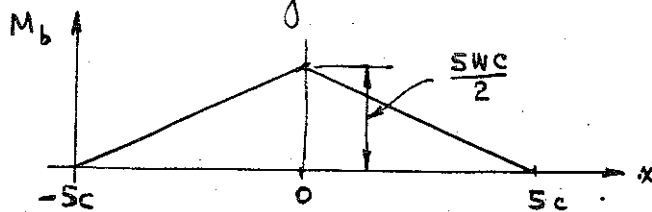
$$\sigma_{78} = 0 ; \sigma_{89} = \frac{W}{A} ; \sigma_{910} = \frac{2W}{A} ; \sigma_{1011} = \frac{2W}{A} ; \sigma_{1112} = \frac{W}{A} ; \sigma_{1213} = 0 ;$$

Bending Theory: Assume simple bending; neglect diagonal members.



$$I_{zz} \approx 2 \left[A \left(\frac{c}{2} \right)^2 \right] = \frac{Ac^2}{2}$$

Moment diagram:

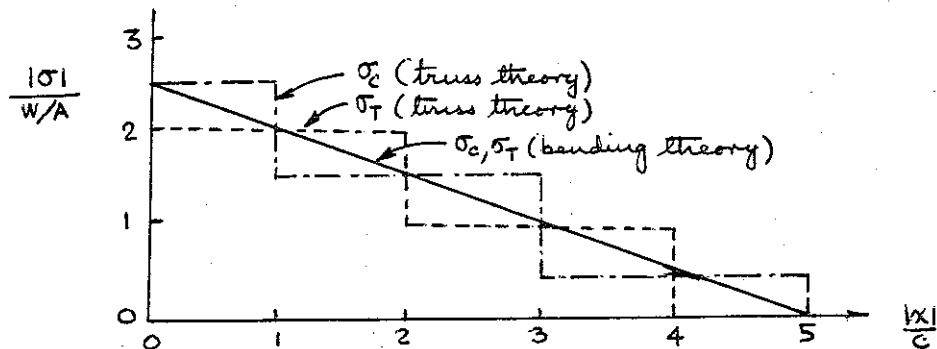


7-30
CONT.

Thus, $M_b = \frac{SWc}{2} \left[1 - \frac{|x|}{5c} \right]$

and $\sigma_T = |\sigma_c| = \frac{M_b \left(\frac{c}{2} \right)}{I_{zz}} = \frac{SW}{2A} \left[1 - \frac{|x|}{5c} \right]$

Comparison :-



Diagonal members transmit shear forces :-

() = () and also the small γ direction forces which arise from the curvature of the beam.

7.31

By symmetry, plane sections remain plane, independent of the stress strain law.

Also, the neutral plane will pass through the section centroid as

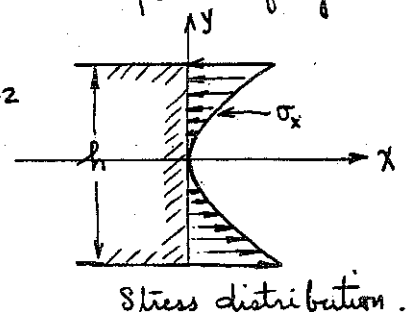
long as the stress strain law is the same for tension and compression. (Note: this holds only for sections symmetric about the z axis.)

Thus, $\epsilon_x = -\frac{y}{R}$ and $\sigma_x = c|\epsilon|^n = \frac{c}{R^n} |y|^n$ negative for $y > 0$ positive for $y < 0$

Using symmetry, $M_b = -2 \int y \sigma_z dA$

$= -2 \int_0^{h/2} by \sigma_x dy = \frac{2bc}{R^n} \int_0^{h/2} (y)(y^n) dy = \frac{2bc}{(n+2)R^n} \left(\frac{h}{2} \right)^{n+2}$

$= \left[\frac{2b}{(n+2)} \left(\frac{h}{2} \right)^2 \right] \left[\frac{c}{R^n} \left(\frac{h}{2} \right)^n \right] = \frac{2b}{(n+2)} \left(\frac{h}{2} \right)^2 (\sigma_x)_{\max}$



$\therefore (\sigma_x)_{\max} = \frac{2M_b(n+2)}{bh^2}$ [check for $n=1$]

7.32

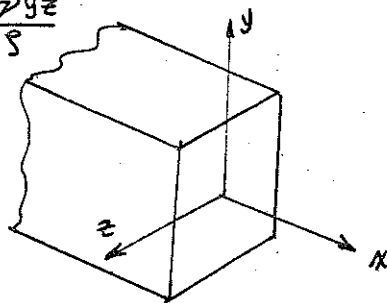
$$u = -\frac{xy}{\rho} ; v = \frac{x^2 + y(y^2 - z^2)}{2\rho} ; w = \frac{zyz}{\rho}$$

$$\epsilon_x = \frac{\partial u}{\partial x} = -\frac{y}{\rho} ; \epsilon_y = \frac{\partial v}{\partial y} = \frac{y}{\rho} ; \epsilon_z = \frac{\partial w}{\partial z} = \frac{y}{\rho}$$

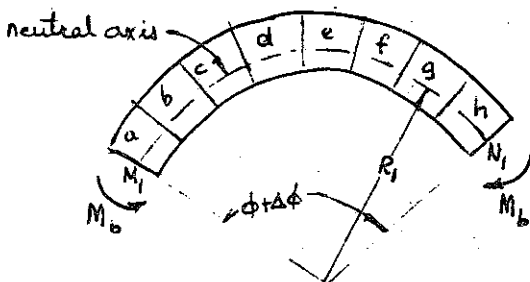
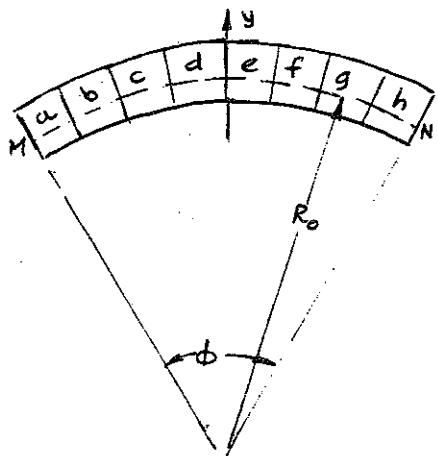
$$\tau_{xy} = \frac{\partial v}{\partial y} + \frac{\partial u}{\partial x} = -\frac{x}{\rho} + \frac{x}{\rho} = 0$$

$$\tau_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} = -\frac{yz}{\rho} + \frac{yz}{\rho} = 0$$

$$\tau_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} = 0 + 0 = 0$$



7.33



Impossible deformed shapes of elements a, b, c...

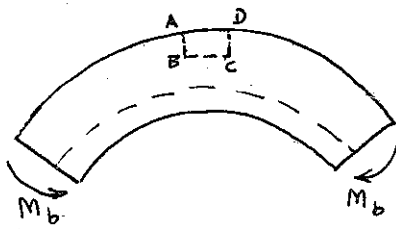
Consider elements a, b, c, d... which are initially circular segments. They are all acted upon by an equal moment and hence deform in a like manner. Also, they must be completely interchangeable even after rotation about the y axis (symmetry). The only possible deformed shape is a circular segment with flat ends which are perpendicular to the curved sides. Hence, radial plane sections remain plane after deformation.

Since the neutral axis has no longitudinal strain (by definition) its length remains constant during deformation.

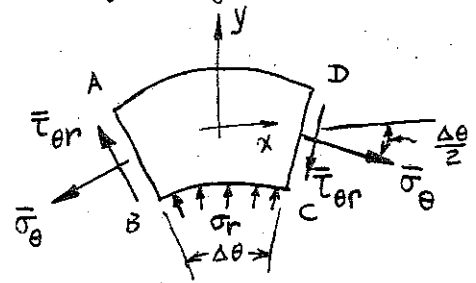
$$\text{i.e. } R_0 \phi = R_1 (\phi + \Delta\phi)$$

$$\therefore \frac{1}{R_1} = \frac{\phi + \Delta\phi}{R_0 \phi} = \frac{1}{R_0} + \frac{\Delta\phi}{R_0 \phi} \quad \text{--- (1)}$$

7-33 (cont)



Free body diagram of ABCD:-



$\sum F_y = 0$ for free body:-

Shear stresses, $\tau_{\theta r}$, on AB and CD cancel each other in the y direction.

Normal stresses, σ_{θ} , have a non-zero y component.

Hence, to balance this, a σ_r must exist unless $\bar{\sigma}_{\theta} = 0$ which cannot be generally true.

Tangential strain distribution:-

Consider a circular arc at distance y from the neutral axis.

Length before deformation = $(R_0 + y)\phi$

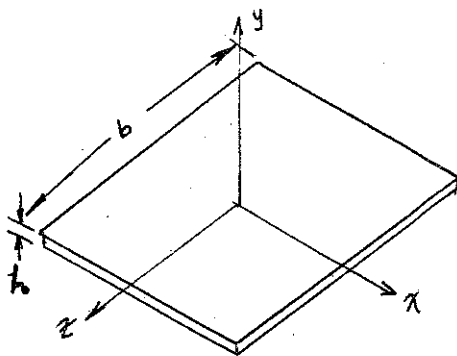
Length after deformation = $(R_1 + y)(\phi + \Delta\phi)$

Thus, tangential strain $\epsilon_{\theta} = \frac{(R_1 + y)(\phi + \Delta\phi) - (R_0 + y)\phi}{(R_0 + y)\phi}$

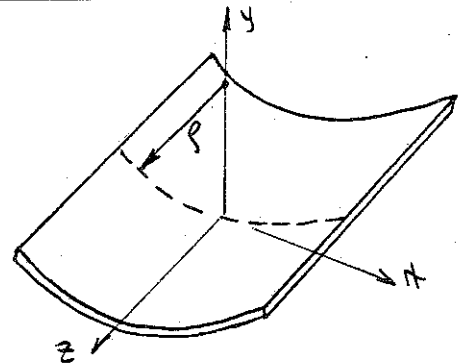
Using (1), this becomes $\epsilon_{\theta} = \frac{R_0 - R_1}{R_1} \left(\frac{y}{R_0 + y} \right)$

which is non linear unless $y \ll R_0$.

7-34

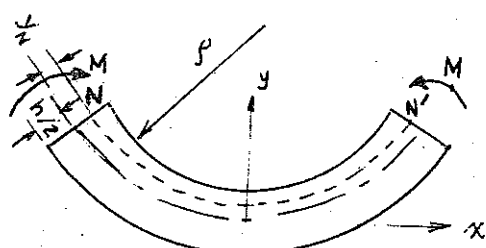


7-19



7-34 (cont)

The experimental observation is that the geometrical quantities (strains etc) are independent of z over the central region. Using this and the symmetry requirement that plane sections remain plane after bending we can write down the following geometrical results:—



Neutral axis is NN' at distance y_N from centerline.

$$\epsilon_x = -\frac{(y-y_N)}{\rho} \quad (1)$$

$$\epsilon_z = \text{Constant} = \epsilon_0 \quad (2)$$

EQUILIBRIUM:

$$\sum F_x = 0 : \int_{-h/2}^{h/2} \sigma_x dy = 0 \quad (3)$$

$$\sum F_z = 0 : \int_{-h/2}^{h/2} \sigma_z dy = 0 \quad (4)$$

$$\sum M_z = 0 : M = -\int_{-h/2}^{h/2} y \sigma_x dy \quad (5)$$

STRESS-STRAIN

$$\epsilon_x = \frac{1}{E} (\sigma_x - \nu \sigma_y - \nu \sigma_z) \quad (6)$$

$$\epsilon_z = \frac{1}{E} (\sigma_z - \nu \sigma_x - \nu \sigma_y) \quad (7)$$

Using (1), (2), (6), (7):

$$\sigma_x = \frac{E}{1-\nu^2} \left[-\frac{(y-y_N)}{\rho} + \nu \epsilon_0 \right]$$

$$\sigma_z = \frac{E}{1-\nu^2} \left[\epsilon_0 - \frac{\nu(y-y_N)}{\rho} \right]$$

Writing $\sum F_x = 0$ and $\sum F_z = 0$ we get:

$$\left(\frac{y_N}{\rho} + \nu \epsilon_0 \right) h = 0$$

$$\left(\epsilon_0 + \frac{\nu y_N}{\rho} \right) h = 0$$

Therefore, unless $\nu = 1$ (impossible), we have:

$$y_N = \epsilon_0 = 0 \quad 7-20$$

7.34 cont

The neutral axis coincides with the centerline and $\epsilon_z = 0$ i.e. we have plane strain, in the x, y plane. (Contrast to plane stress when b/h is small)

$$\therefore \sigma_x = -\frac{E}{1-\nu^2} \frac{y}{s}$$

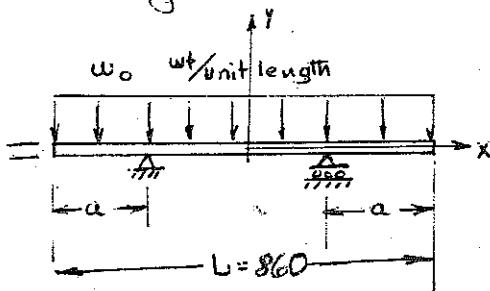
$$\sum M_z = 0 \quad M = -\int_{-h/2}^{h/2} y \left[-\frac{E}{1-\nu^2} \frac{y}{s} \right] dy = \frac{E h^3}{12(1-\nu^2) s}$$

$$\therefore \frac{d\phi}{ds} = \frac{1}{s} = \frac{12(1-\nu^2) M}{E h^3}$$

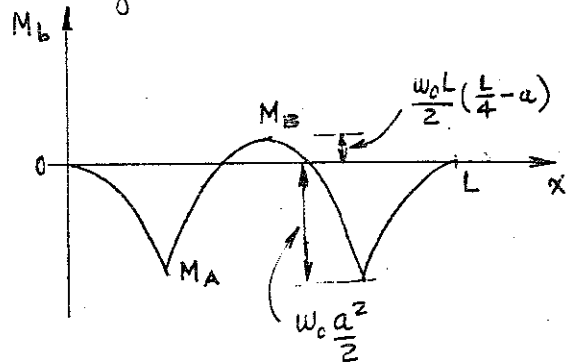
$$\text{and } \sigma_x = -\frac{E y}{1-\nu^2} \left(\frac{1}{s} \right) = \frac{12 M y}{h^3}$$

7.35

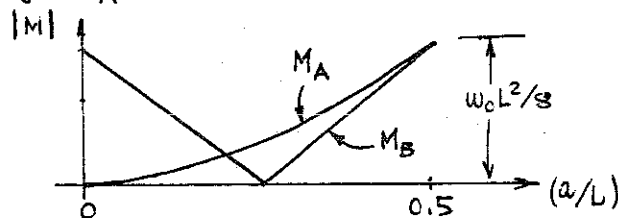
Loading:



Bending Moment:



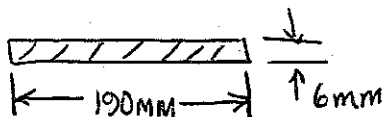
Variation of M_A and M_B :-



Thus minimum bending moment occurs when $M_A = M_B$

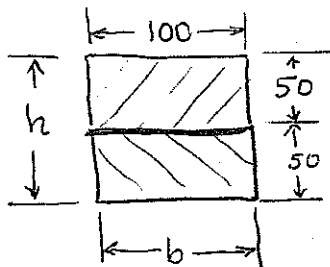
$$(M_b)_{\max} = \frac{w_0 (a)^2}{2} = 15.86 \times 10^3 w_0 ; M_b \text{ PER MM OF WIDTH} = \frac{15.86 \times 10^3 w_0}{100} = 83.47 w_0$$

$$(\sigma_x)_m = \frac{12 M (\frac{h}{2})}{h^3} = \frac{6M}{h^2} = 7 \frac{\text{MN}}{\text{m}^2}, \quad h = 6 \text{ mm}$$



$$\text{gives } w_0 = 500 \text{ N/m}$$

7-36

MOMENT = $3 \times P$ at wall

$$V = -P$$

DESIGN FOR SHEAR FAILURE

$$\tau_{xy} = \frac{QV}{bI_{zz}} ; I_{zz} = \frac{bh^3}{12}$$

$$Q = A\bar{y} = \frac{bh}{2} \times \frac{h}{4} = \frac{bh^2}{8}$$

NO GLUE NEEDED
FOR SECOND
CONFIGURATION

$$\tau_{xy} = \frac{3}{2} \frac{V}{A} = \frac{3}{2} \frac{P}{10^4 \times 10^{-6}} = 1.5 \times 10^2 P \text{ (N/m)}^2$$

7.37

$$\left. \begin{aligned} I_{zz} &= 75 \text{ in}^4 \\ Q_{\text{box}} &= 4.7 \text{ in}^3 \end{aligned} \right\} \text{page 302 of text}$$

$$\tau_{xy} = \frac{VQ}{I_{yy}}$$

a) just above joint :

$$Q = \text{first moment of } 2 \times \frac{1}{4} \text{ strip} \\ = 2\left(\frac{1}{4}\right)(4+1) = 2.5 \text{ in}^3$$

$$\tau_{xy} = \frac{(6000)(2.5)}{75} = \underline{200 \text{ lb/inch}}$$

b) just below joint :

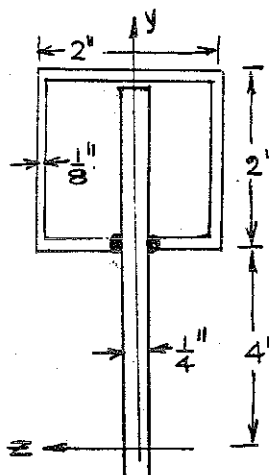
$$Q = (Q \text{ of } 2 \times \frac{1}{4} \text{ strip}) + (Q_{\text{box}}) = 2.5 + 4.7 = 7.2 \text{ in}^3$$

$$\tau_{xy} = \frac{(6000)(7.2)}{75} = \underline{575 \text{ lb/inch}}$$

c) at neutral surface :

$$Q = (Q \text{ of } 6 \times \frac{1}{4} \text{ strip}) + (Q_{\text{box}}) = 6\left(\frac{1}{4}\right)(3) + 4.7 = 9.2 \text{ in}^3$$

$$\tau_{xy} = \frac{(6000)(9.2)}{75} = \underline{740 \text{ lb/inch}}$$



7-22

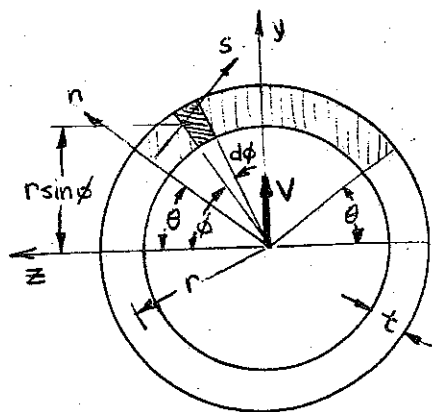
7.38

$$I_{zz} = \pi r^3 t \quad (t/r \ll 1)$$

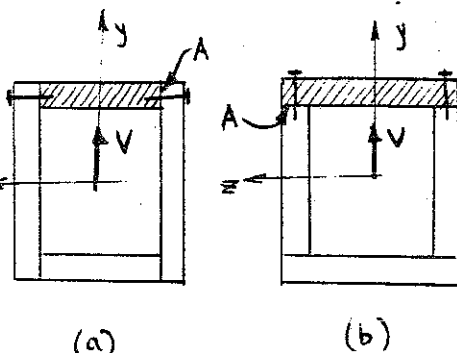
$$Q = \text{first moment of shaded area} \\ = 2 \int_{\theta}^{\pi/2} (r \sin \phi) (t r d\phi) = 2r^2 t \cos \theta$$

$$q_{sx} = q_{xs} = \frac{1}{2} \left(\frac{VQ}{I_{zz}} \right) = \frac{V}{\pi r} \cos \theta$$

(two sides of shaded area)

7.39

The nails transmit the shear flow at the interfaces marked A. The better scheme is the one that requires the lower shear flow to be transmitted, since other things are equal.



$$\text{Shear flow} = \frac{VQ}{I_{zz}} \quad \text{where } Q = \text{first moment of shaded area.}$$

I_{yy} is the same for both schemes. But (a) has the lower Q and hence is better.

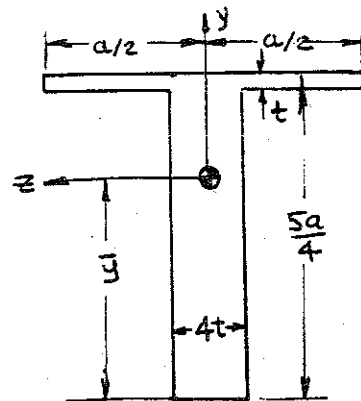
7.40

$$\bar{y} = \frac{(at)(\frac{5a}{4}) + (4t)(\frac{5a}{4})(\frac{5a}{8})}{(at) + (4t)(\frac{5a}{4})} = 0.73a$$

$$(\text{stem } \sigma_x)_{\max} = \frac{M}{I_{zz}} (y_{\max})_{\text{stem}} = \frac{M}{I_{zz}} (0.73a)$$

$$(\text{flange } \sigma_x)_{\max} = \frac{M}{I_{zz}} (y_{\max})_{\text{flange}} = \frac{M}{I_{zz}} (0.52a)$$

$$\therefore \frac{(\text{stem } \sigma_x)_m}{(\text{flange } \sigma_x)_m} = \frac{0.73a}{0.52a} = 1.40$$



Max. τ_{xy} occurs at $y=0$ in the stem.

7-23

7.40 cont

$$Q = \frac{1}{2}(4t)(0.73a)\left(\frac{0.73a}{2}\right) = 1.06 a^2 t$$

flange τ_{zx} is maximum at the junction between flange and stem.

$$Q (\text{half flange}) = \left(\frac{Q}{2}\right)(t)(0.52a) = 0.26 a^2 t$$

$$\text{Thus } \frac{(\text{stem } \tau_{xy})_m}{(\text{flange } \tau_{zx})_m} = \left(\frac{V}{I_{zz}} \frac{1.06 a^2 t}{(4t)}\right) \left(\frac{I_{zz} (t)}{V 0.26 a^2 t}\right) = \underline{\underline{1.02}}$$

Bolts carry shear flow at ends of shaded section.

$$I_{zz} = \frac{1}{12}[(6)(10)^3 - (4)(8)^3] = 330 \text{ in}^4$$

$$Q \text{ for shaded area} = (1)(2)(4.5) = 9 \text{ in}^3$$

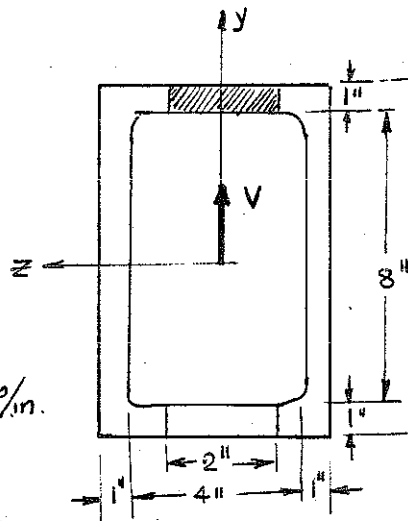
If q_{zx} is the shear flow at each end,

$$2q_{zx} = \frac{VQ}{I_{zz}} = \frac{(10,000)(9)}{330} = 272 \text{ lb/in.}$$

$$\therefore q_{zx} = 136 \text{ lb/in.}$$

Total shear flow for length "s" must be carried by one bolt.

$$\therefore (136)(s) = 400 \quad \text{ie } \underline{\underline{s = 2.95'' \text{ max.}}}$$

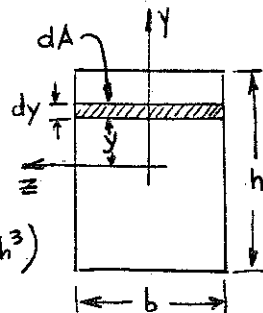


$$\tau_{xy} = \frac{V}{2I_{zz}} \left[\left(\frac{h}{2}\right)^2 - y^2 \right]$$

$$\text{Total shear force} = \int_A \tau_{xy} dA = \int_{y=-h/2}^{h/2} \tau_{xy} b dy$$

$$= \frac{Vb}{2I_{zz}} \int_{-h/2}^{h/2} \left[\left(\frac{h}{2}\right)^2 - y^2 \right] dy = \frac{Vb}{2I_{zz}} \left[h\left(\frac{h}{2}\right)^2 - \frac{1}{3}\left(\frac{h}{2}\right)^3 \right] = \frac{V}{I_{zz}} \left(\frac{1}{12} bh^3 \right)$$

$$\text{And } \frac{1}{12} bh^3 = I_{zz} \quad \therefore \int_A \tau_{xy} dA = V$$



7-43

$$Q = \int_{A_1} y dA = \int_{y_1}^{y_{\max}} by dy = - \int_{y_{\max}}^{y_1} by dy$$

$$\therefore \frac{dQ}{dy_1} = - by_1$$

$$\int_A \tau_{xy} dA = \int_A \frac{VQ}{bI_{zz}} dA = \frac{V}{I_{zz}} \int \frac{Q}{b} (b dy)$$

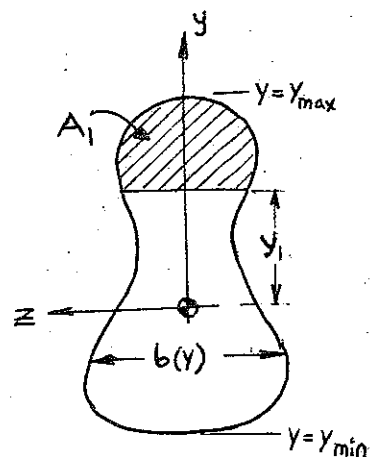
(Note: $b = b(y)$, a function of y)

$$= \frac{V}{I_{zz}} \int_{y_{\min}}^{y_{\max}} Q dy = \frac{V}{I_{zz}} \left[Qy - \int y \frac{dQ}{dy} dy \right]_{y_{\min}}^{y_{\max}}$$

$$= \frac{V}{I_{zz}} \left[Qy + \int by^2 dy \right]_{y_{\min}}^{y_{\max}}$$

The first term in the brackets is zero since $Q = 0$ at $y = y_{\max}$ and $y = y_{\min}$.

$$\therefore \int_A \tau_{xy} dA = \frac{V}{I_{zz}} \int_{y_{\min}}^{y_{\max}} y^2 (b dy) = \frac{V}{I_{zz}} \int_A y^2 dA = \frac{V}{I_{zz}} (I_{zz}) = V$$



7-44

$$\text{Max. bending stress} = (\sigma_x)_m$$

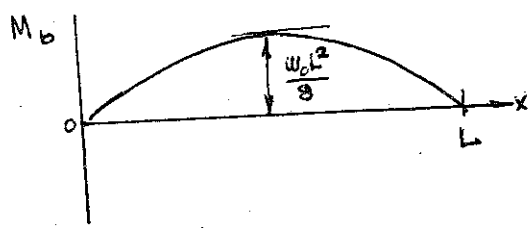
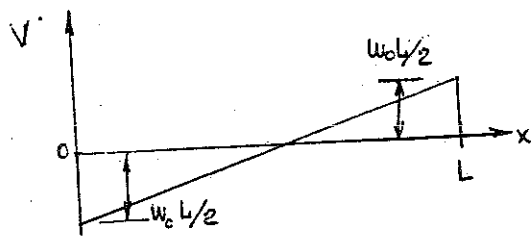
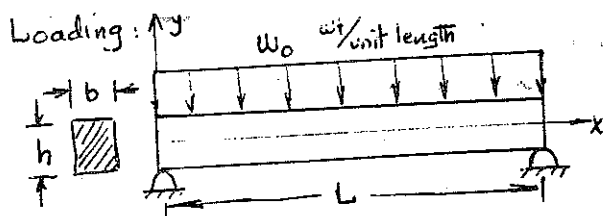
$$= \frac{(M_b)_{\max} \left(\frac{h}{2}\right)}{I_{zz}} = \frac{1}{I_{zz}} \frac{w_0 L^2 h}{16}$$

$$\text{Max. shear stress} = (\tau_{xy})_m$$

$$= \frac{V_{\max}}{b I_{zz}} \int_0^{h/2} y dA = \frac{1}{I_{zz}} \frac{w_0 L h^2}{16}$$

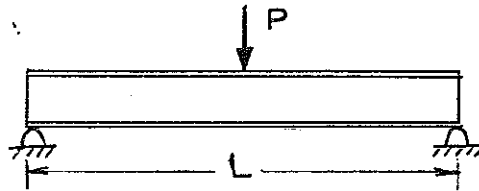
$$\therefore \frac{(\sigma_x)_m}{(\tau_{xy})_m} = \left(\frac{1}{I_{zz}} \frac{w_0 L^2 h}{16} \right) \left(\frac{I_{zz}}{1} \frac{w_0 L h^2}{16} \right)$$

$$= \left(\frac{L}{h} \right)$$

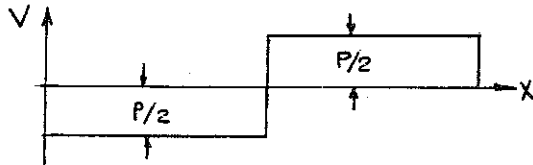


7-25

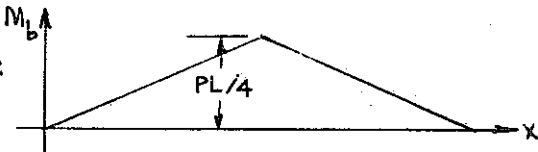
145
Loading:



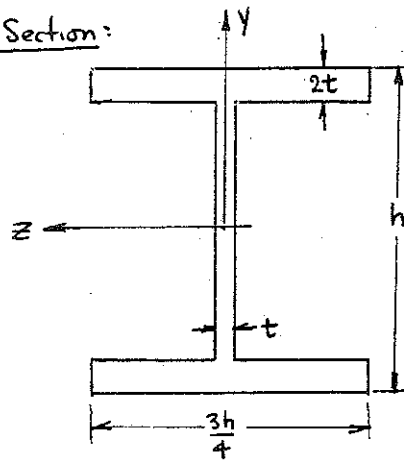
Shear Force:



Bending Moment:



Cross Section:



$$j = h/2$$

$$\int y dA = (t)(\frac{h}{2})(\frac{h}{4}) + (2t)(\frac{3h}{4})(\frac{h}{2}) = \frac{7}{8}th^2$$

$y=0$

$$(\sigma_x)_{\max} = \frac{(M_b)_{\max}(y)_{\max}}{I_{zz}} = \left(\frac{PL}{4}\right)\left(\frac{h}{2}\right)\frac{1}{I_{zz}} = \frac{1}{I_{zz}}\left(\frac{PLh}{8}\right)$$

$$(\tau_{xy})_{\max} = \frac{(V)_{\max}}{I_{zz}} \left[\frac{1}{b} \int y dA \right]_{\max} = \left(\frac{P}{2}\right)\frac{1}{I_{zz}} \left(\frac{1}{t}\right)\frac{7}{8}th^2 = \frac{1}{I_{zz}}\left(\frac{7Ph^2}{16}\right)$$

$$\therefore \frac{(\sigma_x)_{\max}}{(\tau_{xy})_{\max}} = \left(\frac{1}{I_{zz}} \frac{PLh}{8}\right) \left(\frac{I_{zz}}{1} \frac{16}{7Ph^2}\right) = \frac{2}{7}\left(\frac{L}{h}\right)$$

7-46

We will start with equation 7.4 :

$$\epsilon_x = -\frac{d\phi}{ds} y$$

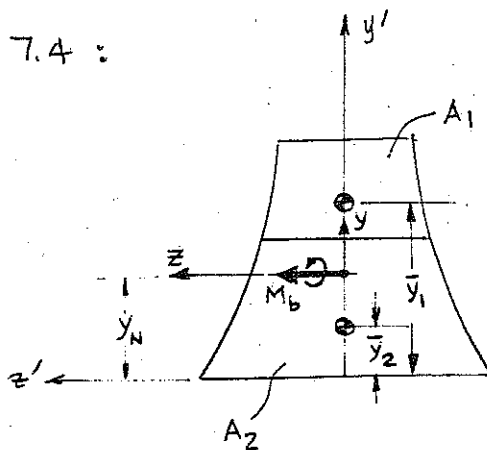
GEOMETRY : $y = y' - y_N$

EQUILIBRIUM :

$$\sigma_y = 0 ; \sigma_z = 0 ;$$

$$\sum F_x = 0 \quad \therefore \int_A \sigma_x dA = 0$$

$$\sum M_z = 0 \quad \therefore M_b + \int_A \sigma_x y dA = 0$$



STRESS-STRAIN : $(\sigma_x)_i = E_i (\epsilon_x)_i = -E_i \left(\frac{d\phi}{ds} \right) y \quad (i=1,2)$

$$\int_A \sigma_x dA = -E_1 \left(\frac{d\phi}{ds} \right) \int_{A_1} (y' - y_N) dA - E_2 \left(\frac{d\phi}{ds} \right) \int_{A_2} (y' - y_N) dA = 0$$

Also, $\int_{A_1} y' dA = \bar{y}_1 A_1$ and $\int_{A_2} y' dA = \bar{y}_2 A_2$

$$\therefore -E_1 \left(\frac{d\phi}{ds} \right) (\bar{y}_1 - y_N) A_1 - E_2 \left(\frac{d\phi}{ds} \right) (\bar{y}_2 - y_N) A_2 = 0$$

$$\text{or } y_N = \frac{E_1 \bar{y}_1 A_1 + E_2 \bar{y}_2 A_2}{E_1 A_1 + E_2 A_2}$$

$$\begin{aligned} M_b &= -\int_A y \sigma_x dA = E_1 \left(\frac{d\phi}{ds} \right) \int_{A_1} (y' - y_N)^2 dA + E_2 \left(\frac{d\phi}{ds} \right) \int_{A_2} (y' - y_N)^2 dA \\ &= E_1 \left(\frac{d\phi}{ds} \right) (I_{zz})_1 + E_2 \left(\frac{d\phi}{ds} \right) (I_{zz})_2 \end{aligned}$$

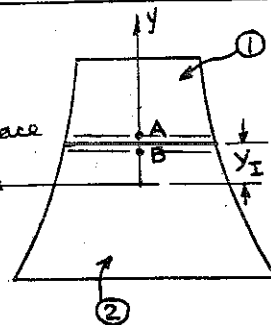
$$\text{or } \frac{d\phi}{ds} = \frac{1}{\rho} = \frac{M_b}{E_1 (I_{zz})_1 + E_2 (I_{zz})_2}$$

$$\text{and } (\sigma_x)_i = -E_i \left(\frac{d\phi}{ds} \right) y = -E_i \frac{M_b y}{E_1 (I_{zz})_1 + E_2 (I_{zz})_2}$$

7.47

Consider the lateral strains, ϵ_z , in two strips of material, one slightly above and the other slightly below the interface between the two materials (A and B in the figure).

Assuming that all stresses but σ_x vanish,



7-27

7.47 cont.

$$(\sigma_x)_A = -\lambda E_1 y_I \quad ; \quad (\sigma_x)_B = -\lambda E_2 y_I \quad ; \quad \text{where } \lambda = \frac{M_b}{E_1(I_{zz})_1 + E_2(I_{zz})_2}$$

$$\therefore (\epsilon_z)_A = \frac{1}{E}(\sigma_z - \nu\sigma_x - \nu\sigma_y) = \lambda \nu_1 E_1 y_I$$

$$(\epsilon_z)_B = \lambda \nu_2 E_2 y_I$$

Thus the lateral strains are, in general, different and if allowed to deform in this manner the cross section would appear as shown alongside :-



This discontinuous displacement violates geometric compatibility and in reality, σ_z stresses will exist and equalise the ϵ_z strains across the interface. Shear stresses τ_{yz} will also exist.

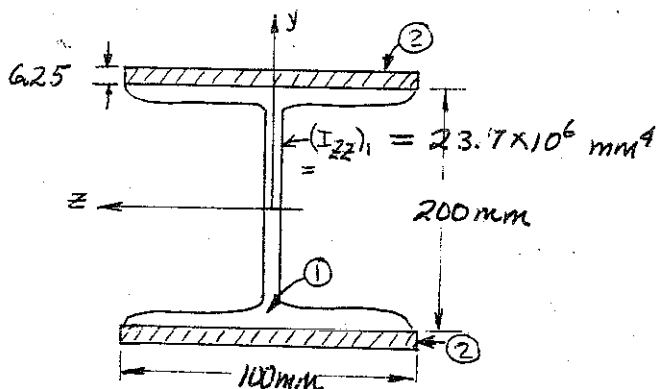
In the special case $E_1 \nu_1 = E_2 \nu_2$, the strains are compatible and only σ_x will exist (to first order).

7.48

Note first that the position of the neutral axis is unaltered by the addition of the plates.

Using the results of prob 7.46 :

$$\begin{aligned} \frac{(K_b)_{\text{stiffened}}}{(K_b)_{\text{unstiffened}}} &= \frac{(\frac{d\phi/ds}{M_b})_{\text{unstiffened}}}{(\frac{d\phi/ds}{M_b})_{\text{stiffened}}} \\ &= \frac{E_1(I_{zz})_1 + E_2(I_{zz})_2}{E_1(I_{zz})_1} = 1 + \frac{E_2}{E_1} \frac{(I_{zz})_2}{(I_{zz})_1} \end{aligned}$$



$$(I_{zz})_2 = 2 \left[\frac{1}{12} (100)(6.25)^3 + 6.25 \times 100 \times (103.125)^2 \right] = 1.329 \times 10^7 \text{ mm}^4$$

$$\frac{(K_b)_{\text{STIFFENED}}}{(K_b)_{\text{UNSTIFFENED}}} = 1 + \frac{205}{78} \frac{1.329 \times 10^7}{2.37 \times 10^7} = 2.47$$

$$\frac{(\sigma_{x \text{ MAX}})_{\text{AL}}}{(\sigma_{x \text{ MAX}})_{\text{STEEL}}} = \frac{E_1 (y_{\text{MAX}})_1}{E_2 (y_{\text{MAX}})_2} = \frac{78 \times 100}{205 \times 103} = 0.369$$

7-49

USE RESULTS OF PROB 7.46

$$(I_{zz})_1 = \frac{\pi}{64} [148^4 - (42)^4] = 1.078 \times 10^{-7} \text{ m}^4$$

$$(I_{zz})_2 = \frac{\pi}{64} [142^4 - 130^4] = 1.13 \times 10^{-7} \text{ m}^4$$

$$E_1 (I_{zz})_1 + E_2 (I_{zz})_2 = 200 (I_{zz})_1 + 78 (I_{zz})_2 = 3.037 \times 10^4 \text{ N} \cdot \text{m}^2$$

$$\text{MAX STRESS IN STEEL} = \frac{M_b E_1 \times 24}{30.37 \times 10^4} = 1.58 \times 10^5 M_b$$

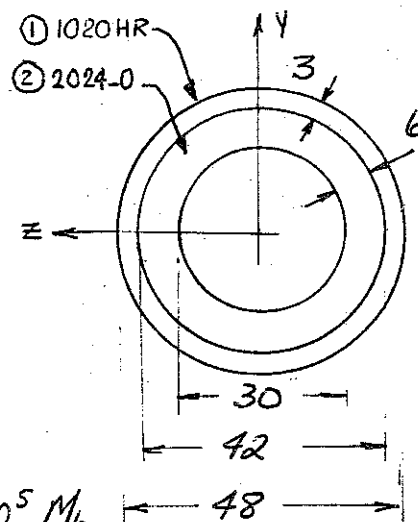
$$\text{MAX STRESS IN ALUMINUM} = 0.539 \times 10^5 M_b$$

$$Y_{\text{STEEL}} = 259 \text{ MN/m}^2 ; Y_{\text{AL}} = 98 \text{ MN/m}^2$$

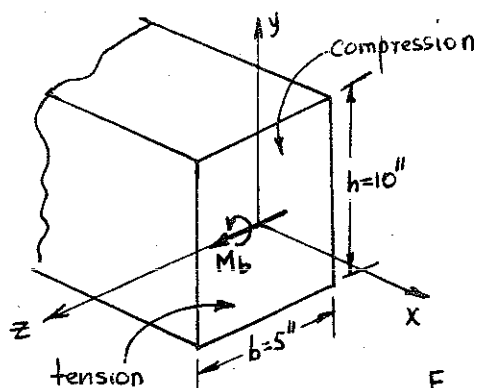
FOR YIELDING

$$M_b = 1.638 \times 10^3 \text{ N} \cdot \text{m} \quad M_b = 1.818 \times 10^3 \text{ N} \cdot \text{m}$$

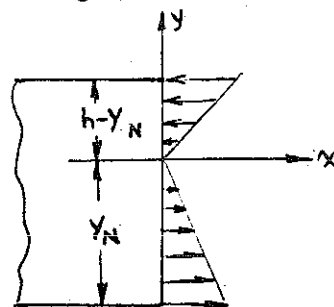
$$\text{MAXIMUM BENDING MOMENT } 1.818 \times 10^3 \text{ N} \cdot \text{m}$$



7-50



Elastic stress distribution :-



$$E_T = E_{\text{tension}} = 7.5 \times 10^6 \text{ psi} ; Y_T = 10,000 \text{ psi}$$

$$E_C = E_{\text{comp}} = 30 \times 10^6 \text{ psi} ; Y_C = 20,000 \text{ psi}$$

Y_N = height of neutral axis ; ρ = radius of curvature.

GEOMETRY : $E_x = -y/\rho$

EQUILIBRIUM : $\sigma_y = \sigma_z = 0 ; \sum F_x = \int_{-Y_N}^{h-Y_N} b \sigma_x dy = 0$

7-29

7.50 cont

$$\Sigma M_z = M_b + \int_{-Y_N}^{h-Y_N} b \sigma_x y dy = 0$$

STRESS-STRAIN . Compression : $\sigma_x = E_c \epsilon_x$

Tension : $\sigma_x = E_T \epsilon_x$

YIELD CRITERION : Compression : $|\sigma_x| = Y_c$

Tension : $|\sigma_x| = Y_T$

Thus :

$$\Sigma F_x = -\frac{E_c}{9} \int_0^{h-Y_N} b y dy - \frac{E_T}{9} \int_{-Y_N}^0 b y dy = 0$$

$$\text{i.e. } -\frac{E_c b}{9} \frac{1}{2} (h-Y_N)^2 + \frac{E_T b}{9} \frac{1}{2} Y_N^2 = 0 \quad \text{giving } \underline{Y_N = \frac{2h}{3}}$$

Using this value of Y_N .

$$\Sigma M_z = M_b - \frac{E_c}{9} \int_0^{h/3} b y^2 dy + \frac{E_T}{9} \int_{-2h/3}^0 b y^2 dy = 0$$

$$\therefore M_b = \frac{bh^3}{81} [E_c + 8E_T]$$

$$\text{Max. compressive stress} = \sigma_c = E_c \left(\frac{h/3}{9}\right) = E_c \left(\frac{h}{3}\right) \frac{81 M_b}{bh^3 [E_c + 8E_T]}$$

$$\text{Max. tensile stress} = \sigma_T = \frac{E_T (2h/3)}{9} = E_T \left(\frac{2h}{3}\right) \frac{81 M_b}{bh^3 [E_c + 8E_T]}$$

$$\text{Thus } M_b = \frac{bh^2}{27} \sigma_c \left[1 + 8 \frac{E_T}{E_c}\right] = \frac{bh^2}{27} \frac{\sigma_T}{2} \left[\frac{E_c}{E_T} + 8\right]$$

$$\frac{bh^2}{27} = \frac{(5)(100)}{27} = 18.5 ; \left(1 + 8 \frac{E_T}{E_c}\right) = \left(1 + \frac{8(7.5)}{30}\right) = 3$$

$$\left(\frac{E_c}{E_T} + 8\right) = \left(\frac{30}{7.5} + 8\right) = 12$$

$$\text{Thus for } |\sigma_c| = Y_c : M_b = (18.5)(20,000)(3) = 1.11 \times 10^6 \text{ in-lb.}$$

$$\text{for } \sigma_T = Y_T : M_b = (18.5) \left(\frac{10,000}{2}\right) (12) = 1.11 \times 10^6 \text{ in-lb.}$$

$$\text{Thus } (M_b)_{\max} = \underline{1.11 \times 10^6 \text{ in-lb.}}$$

[Note: The problem may also be solved by using the results of prob. 7.46]

7-51

GEOMETRY : $\epsilon_x = -\frac{y}{\rho}$

EQUILIBRIUM :

$\sigma_y = \sigma_z = 0$

$\sum F_x = \int \sigma_x dA = 0$

$\sum M_z = M_b + \int \sigma_x y dA = 0$

STRESS-STRAIN :

Concrete : Compression : $\sigma_x = E_c \epsilon_x$
Tension : $\sigma_x = 0$

Steel : $\sigma_x = E_s \epsilon_x$

We will take a mean value of strain (and stress) for the steel.

$$\sum F_x = - \int_0^{kd} \frac{E_c y}{\rho} b dy + E_s \frac{(1-k)d}{\rho} A_s = 0$$

i.e.
$$E_s A_s (d-kd) - E_c b \frac{(kd)^2}{2} = 0 \quad \text{--- (1)}$$

$$\sum M_z = M_b - \int_0^{kd} \frac{E_c y}{\rho} b y dy - E_s \frac{(1-k)d}{\rho} A_s (1-k)d = 0$$

giving
$$\frac{1}{\rho} = \frac{M_b}{\frac{1}{3} E_c b k^3 d^3 + E_s A_s (1-k)^2 d^2}$$

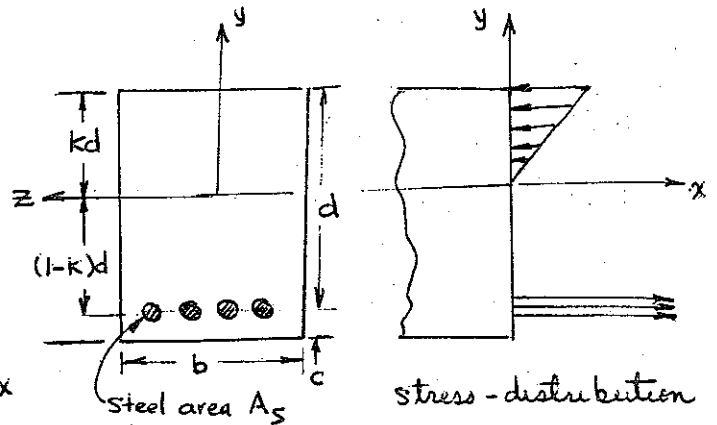
$$\sigma_{\text{steel}} = \sigma_s = E_s \frac{(1-k)d}{\rho} = \frac{M_b E_s (1-k)d}{\frac{1}{3} E_c b k^3 d^3 + E_s A_s (1-k)^2 d^2} = \frac{M_b}{A_s d \left[\frac{1}{3} \frac{E_c b k^3 d}{A_s E_s (1-k)} + (1-k) \right]}$$

From (1)
$$\frac{E_c b d}{E_s A_s} = \frac{2(1-k)}{k^2} \quad \therefore \frac{1}{3} \frac{E_c b k^3 d}{A_s E_s (1-k)} = \frac{2}{3} k$$

and
$$\sigma_s = \frac{M_b}{A_s d \left[\frac{2k}{3} + 1 - k \right]} = \frac{M_b}{A_s d \left[1 - \frac{k}{3} \right]}$$

$$(\sigma_{\text{concrete}})_m = (\sigma_c)_m = E_c \frac{(kd)}{\rho} = \sigma_s \frac{kd}{(1-k)d} \frac{E_c}{E_s} = \sigma_s \frac{E_c}{E_s} \left(\frac{k}{1-k} \right)$$

From (1)
$$\frac{E_c}{E_s} = \frac{2A_s(1-k)}{bdk^2} \quad \therefore (\sigma_c)_m = \frac{M_b}{A_s d \left[1 - \frac{k}{3} \right]} \frac{2A_s(1-k)}{bdk^2} \frac{k}{1-k} = \frac{2M_b}{bd^2 k \left[1 - \frac{k}{3} \right]}$$



7.52

Referring to prob. 7.51, the equation for k is:

$$E_s A_s (1-k) - \frac{1}{2} E_c b k^2 d = 0$$

$$\text{or } k^2 + \frac{2E_s A_s}{E_c b d} k - \frac{2E_s A_s}{E_c b d} = 0$$

$$\frac{2E_s A_s}{E_c b d} = \frac{(2)(30 \times 10^6)(2.2)}{(1.5 \times 10^6)(9)(16)} = 0.61$$

$$\therefore k^2 + 0.61k - 0.61 = 0 \quad \text{giving } \underline{k = 0.475}$$

For $\sigma_s = 20,000$ psi $M_b = 20,000 A_s d (1 - \frac{k}{3})$
 $= 20,000(2.2)(16)(0.842) = 592,000$ in-lb.

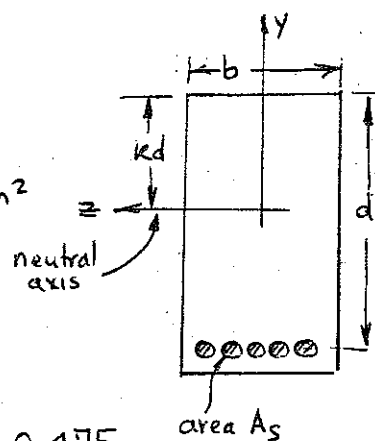
For $(\sigma_c)_{\max} = 1350$ psi, $M_b = (1350) \frac{1}{2} b d^2 k (1 - \frac{k}{3})$
 $= (1350)(\frac{1}{2})(9)(256)(0.475)(0.842) = 625,000$ in-lb.

$$(M_b)_{\max} = 592,000 \text{ in-lb}$$

$$b = 9"$$

$$d = 16"$$

$$A_s = 2.2 \text{ in}^2$$

7.53

Refer to figure for prob 7.51.

$$\sigma_s = \frac{E_s (1-k) d}{s} ; (\sigma_c)_{\max} = \frac{E_c k d}{s} \quad \therefore \frac{\sigma_s}{(\sigma_c)_{\max}} = \frac{1-k}{k} \left(\frac{E_s}{E_c} \right)$$

$$\frac{20,000}{1350} = \frac{1-k}{k} \left(\frac{30 \times 10^6}{1.5 \times 10^6} \right) \quad \text{giving } \underline{k = 0.575}$$

From eqn (1) of prob 7.33 : $A_s = \frac{1}{2} \frac{E_c}{E_s} (b d) \frac{k^2}{1-k}$

$$= \frac{1}{2} \left(\frac{1.5 \times 10^6}{30 \times 10^6} \right) (9)(16) \frac{(0.332)}{(0.425)} = \underline{2.8 \text{ in}^2}$$

i.e. $(5) \frac{\pi}{4} d_s^2 = 2.8$ where d_s = diameter of steel bars

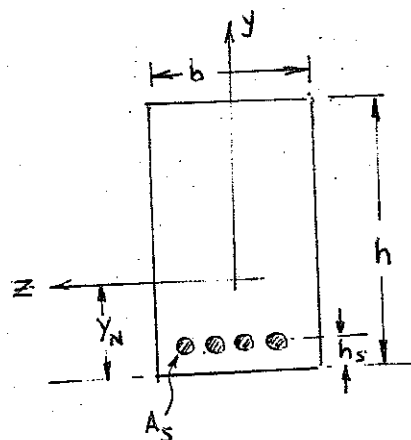
giving $\underline{d_s = 0.84 \text{ in}}$

$$M_b = \sigma_s A_s d (1 - \frac{k}{3}) = 20,000(2.8)(16)(1 - \frac{0.575}{3}) = \underline{728,000 \text{ in-lb.}}$$

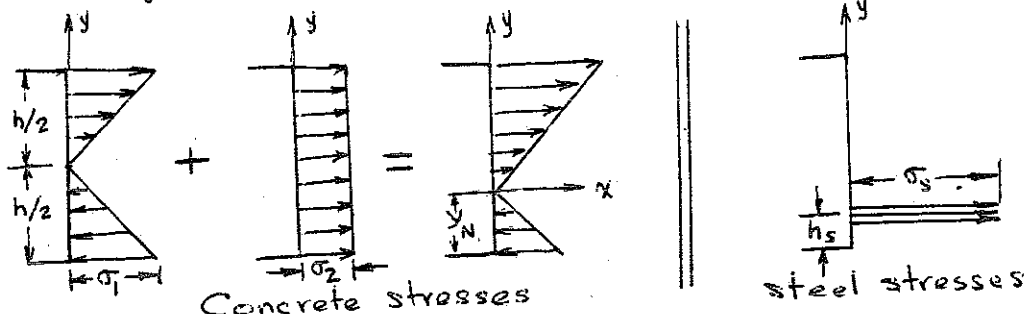
7.54

a) PRESTRESSING:

When the wires are released the beam will bend to a radius of curvature ρ_1 . The strain and stress distribution in the concrete will be linear with the neutral axis located at height y_N .



The stress distribution in the concrete may be decomposed into a symmetrical bending stress and a uniform stress. Thus -



Note that the bending stress has zero resultant and that the uniform stress has zero moment about the centerline.

EQUILIBRIUM : $\sum F_x = \sigma_2(bh) + \sigma_s A_s = 0$ ————— (1)

$\sum M_z = -2 \left[\left(\frac{\sigma_1}{2} \right) \left(\frac{h}{2} \right) (b) \right] \left[\frac{2}{3} \left(\frac{h}{2} \right) \right] + \sigma_s A_s \left(\frac{h}{2} - h_s \right) = 0$ — (2)

From (1) : $\sigma_2 = - \frac{\sigma_s A_s}{bh}$

From (2) : $\sigma_1 = \frac{6 \sigma_s A_s}{bh^2} \left(\frac{h}{2} - h_s \right)$

$A_s = 76 \frac{\pi}{4} (0.15)^2 = 1.34 \text{ in}^2$; $b = 9''$; $h = 18''$; $h_s = 6''$;

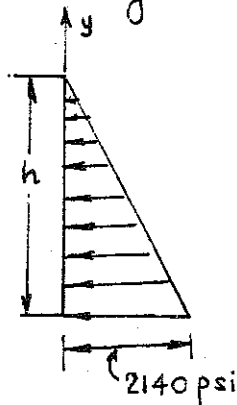
$\sigma_s = 130,000 \text{ psi}$; giving

$\sigma_2 = \frac{-(130,000)(1.34)}{(9)(18)} = -1070 \text{ psi}$

$\sigma_1 = \frac{6(130,000)(1.34)(9-6)}{(9)(18)(18)} = 1070 \text{ psi}$

(7.54 cont.)

The resulting stress distribution in the concrete is:-



[Note: tensile stresses will be absent when

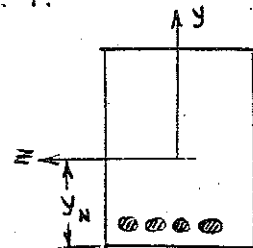
$$\frac{2h}{3} \geq h_s \geq \frac{h}{3}]$$

b) BENDING: We will superimpose on the above stress distribution, that due to bending. In calculating the bending stresses we will not assume that the concrete cannot carry tensile stresses since these will eventually be offset by the compressive prestresses. This should be checked after the solution has been completed.

Bending stresses. Formulae of prob. 7. ^{4b}

$$y_N = \frac{E_c \bar{y}_c A_c + E_s \bar{y}_s A_s}{E_c A_c + E_s A_s}$$

$$\frac{1}{\rho_b} = \frac{M_b}{E_c (I_{zz})_c + E_s (I_{zz})_s}$$



where $\frac{1}{\rho_b} = \text{change in curvature due to } M_b$

here, $\bar{y}_c = 9"$; $A_c = (9)(18) = 162 \text{ in}^2$

$\bar{y}_s = 6"$; $A_s = 1.34 \text{ m}^2$

$E_c = 1.5 \times 10^6 \text{ psi}$; $E_s = 30 \times 10^6 \text{ psi}$

$$\therefore y_N = \frac{(1.5 \times 10^6)(9)(162) + (30 \times 10^6)(6)(1.34)}{(1.5 \times 10^6)(162) + (30 \times 10^6)(1.34)} = \underline{8.5"}$$

$$(I_{zz})_c = \left[\frac{1}{12} (9)(18)^3 + (9)(18)(9-8.5)^2 \right] = 4170 \text{ in}^4$$

$$(I_{zz})_s = \left[0 + (1.34)(8.5-6)^2 \right] = 8.4 \text{ in}^4$$

7.54 cont

$$E_c(I_{zz})_c + E_s(I_{zz})_s = 10^6 [(1.5)(4170) + (30)(8.4)] = 6512 \times 10^6 \text{ lb-in}^2$$

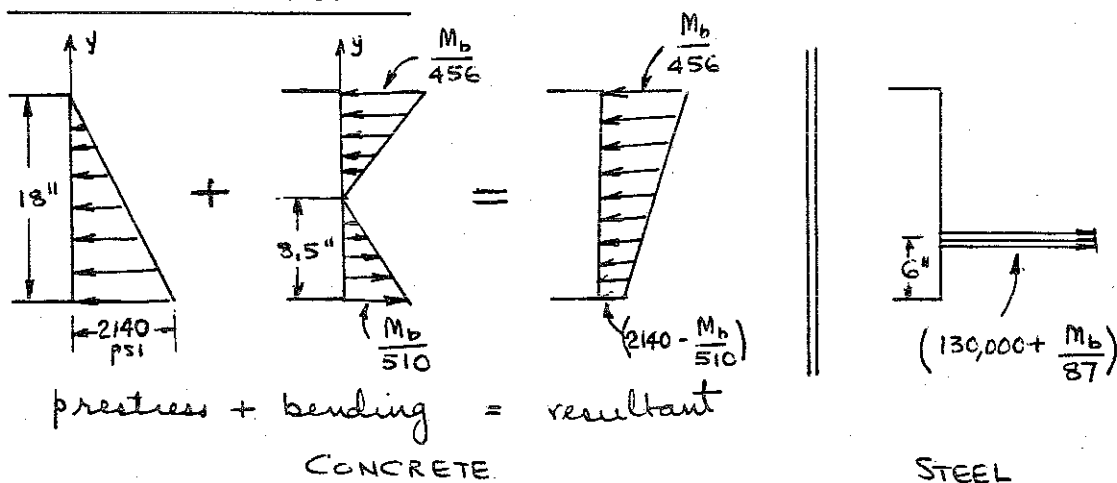
$$(\sigma_x)_{\text{concrete}} = - \frac{M_b(1.5 \times 10^6)y}{6512 \times 10^6} \quad \text{where } y \text{ is measured from the neutral axis}$$

$$\text{Max. compressive stress} = \frac{M_b(1.5 \times 10^6)(9.5)}{6512 \times 10^6} = \frac{M_b}{456}$$

$$\text{Max. tensile stress} = \frac{M_b(1.5 \times 10^6)(8.5)}{6512 \times 10^6} = \frac{M_b}{510}$$

$$\text{Bending stress in steel} = \frac{M_b(30 \times 10^6)(8.5 - 6)}{6512 \times 10^6} = \frac{M_b}{87}$$

c) RESULTANT STRESSES



REQUIREMENTS

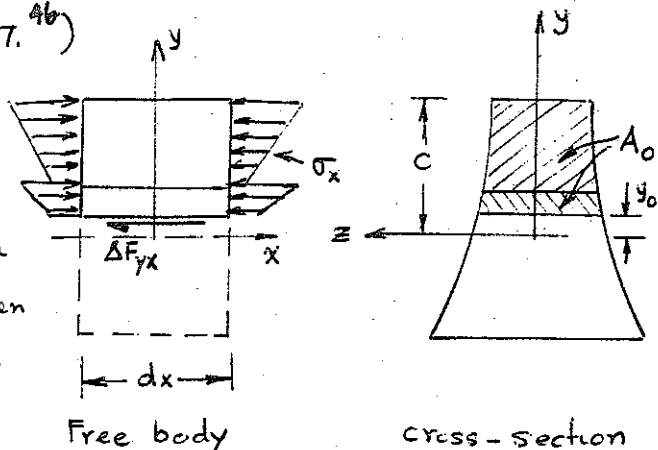
- 1) No tension in concrete : $2140 - \frac{M_b}{510} > 0 \quad \therefore M_b < 1,090,000 \text{ in-lb}$
- 2) Concrete compressive stress $< 2250 \text{ psi}$: $\frac{M_b}{456} < 2250 \quad \therefore M_b < 1,030,000 \text{ in-lb}$
- 3) Steel tensile stress $< 150,000 \text{ psi}$: $130,000 + \frac{M_b}{87} < 150,000 \quad \therefore M_b < 1,740,000 \text{ in-lb}$

$$(M_b)_{\text{max}} = 1,030,000 \text{ in-lb}$$

7.55

$$\sigma_x = \frac{-E_i M_b y}{E_1(I_{zz})_1 + E_2(I_{zz})_2} \quad (\text{prob 7.46})$$

As in section 7.6, we consider the equilibrium of a portion of the beam of length dx , between $y = y_0$ and $y = c$. (free body shown).



Eqn 7.18 (see 7.6) : $\Delta F_{yx} = \left[\int_{A_0} \sigma_x dA \right]_{x+dx} - \left[\int_{A_0} \sigma_x dA \right]_x$

$$\begin{aligned} \text{or } \frac{dF_{yx}}{dx} &= \frac{d}{dx} \int_{A_0} \sigma_x dA = \frac{d}{dx} \left[-\frac{M_b}{E_1(I_{zz})_1 + E_2(I_{zz})_2} \int_{A_0} E y dA \right] \\ &= -\left(\frac{dM_b}{dx} \right) \left(\frac{1}{E_1(I_{zz})_1 + E_2(I_{zz})_2} \right) \left(\int_{A_0} E y dA \right) \end{aligned}$$

Now $-\frac{dM_b}{dx} = V$ and $\tau_{xy} = \tau_{yx} = \frac{q_{yx}}{b} = \frac{1}{b} \frac{dF_{yx}}{dx}$

$$\therefore \tau_{xy} = \tau_{yx} = \frac{V}{b[E_1(I_{zz})_1 + E_2(I_{zz})_2]} \int_{A_0} E y dA$$

56

REFER TO FIGURE, PROB 7.48

$$(I_{zz})_1 = 2.37 \times 10^7 \text{ mm}^4 ; (I_{zz})_2 = 1.33 \times 10^7 \text{ mm}^4 ;$$

$$E_1(I_{zz})_1 + E_2(I_{zz})_2 = 10^{10} [78 \times 2.37 + 205 \times 1.33] = 457.51 \times 10^{10} \text{ N mm}^2$$

FROM PROB 7.55 : THE INTERFACIAL SHEAR FLOW IS GIVEN BY

$$q_{yx} = \frac{V}{[E_1(I_{zz})_1 + E_2(I_{zz})_2]} \int_{A_0} E y dA \quad \text{WHERE THE INTEGRATION IS TO BE CARRIED OUT OVER THE STEEL PLATE}$$

$$\int_{A_0} E y dA = 205 \times 10^3 \times 103 \times 100 \times 6.25 = 1.32 \times 10^{10}$$

$$q_{yx} = \frac{27 \times 1.32 \times 10^{10}}{4.57 \times 10^{12}} = 78 \text{ N/mm} \quad \text{THIS SHEAR FLOW MUST BE TRANSMITTED BY THE RIVETS}$$

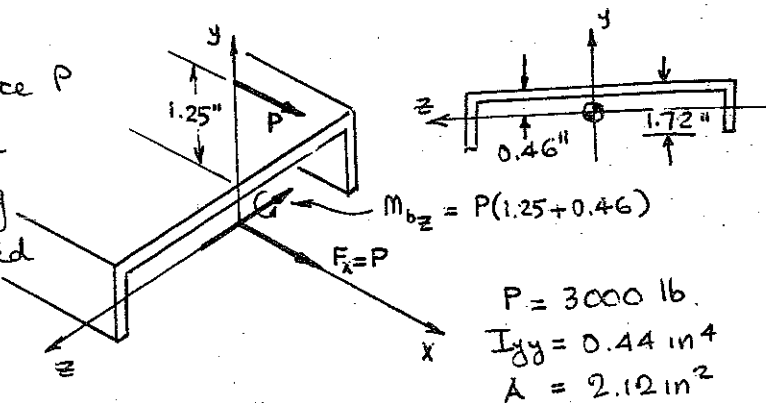
TWO ROWS OF RIVETS, WITH LONGITUDINAL SPACING d

$$\text{LOAD/RIVET} = \frac{q_{yx} d}{2} = 78$$

$$d = \frac{2 \times 2.2 \times 10^3}{78} = 56.4 \text{ mm}$$

7.57

The clamping force P is equivalent to a force $F_x (=P)$ acting through the centroid with a moment M_{bz} , as shown.



$$M_{bz} = 3000(1.25 + 0.46) = 5150 \text{ in-lb.}$$

$$\text{Force } F_x \text{ produces uniform tension} = (\sigma_x)_0 = \frac{3000}{2.12} = 1410 \text{ psi}$$

$$\text{Moment } M_b \text{ produces max. compression at lower edge of channel} = \frac{5150(1.72 - 0.46)}{0.44} = 14,700 \text{ psi}$$

$$\text{and max. tension at upper edge} = \frac{5150(0.46)}{0.44} = 5200 \text{ psi}$$

$$\therefore \text{Max. compression} = 14700 + 1410 = 13290 \text{ psi; Max tension} = 5200 + 1410 = 6610 \text{ psi}$$

$$\text{and therefore } |\sigma_x|_{\max} = \underline{\underline{13,290 \text{ psi}}}$$

7.58 From prob 7.23 :

$$(M_{bz})_{\max} = \frac{\pi r^2 L^2 \gamma}{8}$$

For a thin walled section

$$I_{yy} \cong \pi r^3 t$$

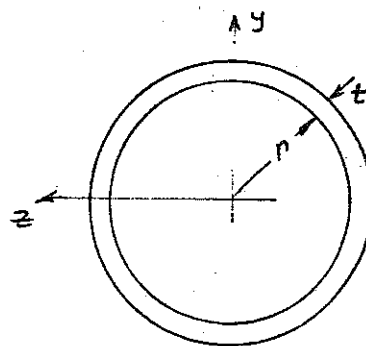
$$\therefore (\sigma_b)_{\max} = \left(\frac{\pi r^2 L^2 \gamma}{8} \right) \left(\frac{1}{\pi r^3 t} \right) (r) = \frac{\gamma L^2}{8t}$$

pressure p causes axial tensile stress = $\frac{pr}{2t}$

$$\therefore \text{max. axial tensile stress when pressurised} = \frac{\gamma L^2}{8t} + \frac{pr}{2t}$$

$$\text{max. axial tensile stress with vent open} = \frac{\gamma L^2}{8t}$$

$$\frac{\gamma L^2}{8t} + \frac{pr}{2t} = 2 \left(\frac{\gamma L^2}{8t} \right) \quad \therefore p = \frac{\gamma L^2}{4r}$$



7.59 We will assume that the cutting forces can be replaced by point loads acting at 4.95mm radius (mid-depth of cut).

Moments and torques at the collet :-

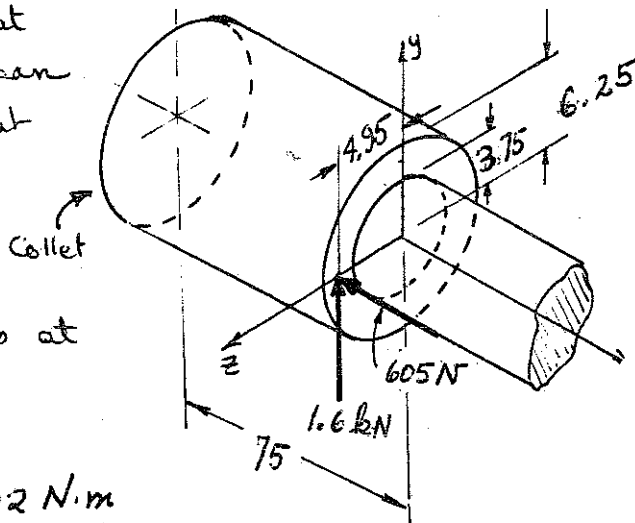
a) DUE TO 1.6 kN FORCE:

$$M_{Lz} = -(1.6) \times 4.95 = -7.92 \text{ N.m}$$

$$M_{bz} = 1.6 \times 7.5 = 120 \text{ N.m}$$

b) DUE TO 605 N FORCE

$$M_{by} = -605 \times 4.95 = -3.0 \text{ N.m}$$



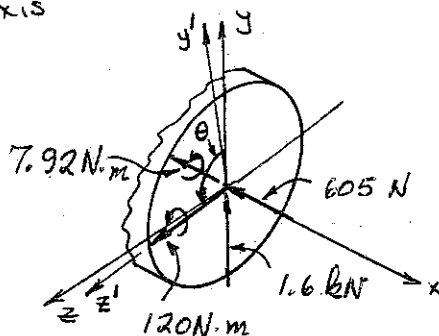
7-59 cont

The moments M_{by} and M_{bz} can be combined to give a single moment $M_b = (M_{by}^2 + M_{bz}^2)^{1/2}$ at an angle $\theta = \tan^{-1}(M_{bz}/M_{by})$ to the y axis

$$M_b = [(3.0)^2 + (120)^2]^{1/2} = 120 \text{ N}\cdot\text{m}$$

$$\theta = \tan^{-1}(120/-3) \approx 91.4^\circ$$

Resultant moments, forces, torques are shown in the figure alongside.



Choose new coordinate axes y' and z' as shown, so that the resultant bending moment is along z' .

M_b now acts along a principal axis of inertia.

$$I_{z'z'} = \frac{\pi}{4} (6.25)^4 = 1198 \text{ mm}^4 ; I_x = 2 I_{z'z'} = 2397 \text{ mm}^4 ; A = 122.7 \text{ mm}^2$$

STRESSES

1) 605 N FORCE: $\sigma_x = -\frac{605}{122.7 \times 10^{-6}} = -4.93 \text{ MN/m}^2$ (UNIFORM)

2) 1.6 kN FORCE: $\tau_{xy'} = \frac{1.6 Q}{b I_{z'z'}}$; THIS IS A FUNCTION OF y' AND IS ZERO AT TOP + BOTTOM

$$\text{MAX at } y=0 ; Q_{\text{max}} = \frac{2}{3} r^3$$

$$(\tau_{xy})_{\text{max}} = \frac{1.6 \times 0.667 \times (6.25)^3 \times 10^{-9}}{13 \times 10^{-3} \times 1198 \times 10^{-12}} = 16.7 \text{ MN/m}^2$$

3) M_b : $\sigma_x = \frac{M_b y'}{I_{z'z'}}$; THIS IS MAXIMUM AT $y = \pm 6.25$; ZERO AT $y=0$

$$(\sigma_x)_{\text{max}} = \frac{120 \times 6.25}{1198 \times 10^{-12}} = 626 \text{ MN/m}^2$$

4) M_t : $\tau = \frac{M_t r}{I_x}$ THIS IS MAXIMUM AT $r = 6.25$ AND ZERO AT $r=0$

$$\tau_{\text{max}} = \frac{7.92 \times 6.25 \times 10^{-3}}{2.397 \times 10^{-9}} = 20.6 \text{ MN/m}^2$$

7.59 cont

Resultant stresses: linear superposition. The process of writing the various stresses as functions of y' and z' ; adding these, finding the max. shear stress, and finally maximising this with respect to y' and z' is an extremely tedious procedure.

To simplify the problem we note that the bending stress is so much larger than the other stresses that the position of max. shear stress will approximately coincide with that of max. bending stress.

Thus, at $y' = 6.25$; $z' = 0$:-

$$\sigma_x = 626 - 4.9 = 621.1 \text{ MN/m}^2$$

$$\tau_{xz} = 20.6 \text{ MN/m}^2$$

$$\text{and } \tau_{\max} = \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + (\tau_{xz})^2} = 311 \text{ MN/m}^2$$

7.60

Loading is shown for one side frame.

Equilibrium : $R_1 \approx 315 \text{ N}$
 $R_2 \approx 250 \text{ N}$

Stresses are due to bending.

Find max. bending moment:

$$AB : M_b \approx 180 \times 0.350 = 63 \text{ N}\cdot\text{m}$$

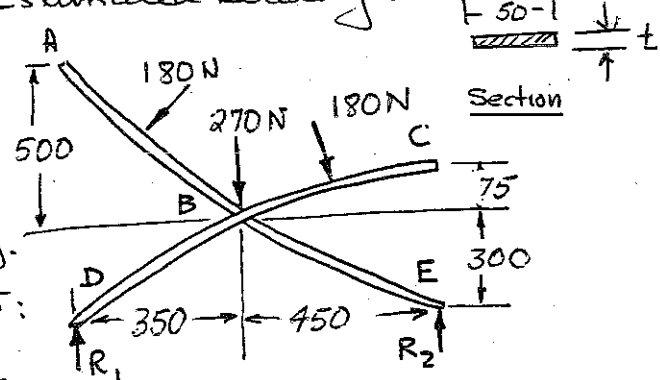
$$BC : M_b \approx 180 \times 0.250 = 45 \text{ N}\cdot\text{m}$$

$$BD : M_b \approx 315 \times 0.350 = 110 \text{ N}\cdot\text{m}$$

$$BE : M_b \approx 250 \times 0.45 = 112 \text{ N}\cdot\text{m} \quad \therefore (M_b)_{\max} \approx 112 \text{ N}\cdot\text{m}$$

$$\text{DESIGN STRESS} = \frac{\gamma}{2} \text{ say } = 294 \text{ MN/m}^2 \quad \therefore 294 \times 10^6 = \frac{112 (t/2)}{\frac{1}{2} t^3 50 \times 10^{-3}}$$

Estimated Loading :-



7-40

$$t = 2.76 \text{ mm}$$

A hand-drawn diagram of a stepped profile. The profile has a total width of 6.25. The left side has a vertical height of 4.86. The top horizontal segment has a width of 1.39. The right side has a vertical height of 5.45. The bottom horizontal segment has a width of 1.60. The total height of the profile is 10.9. The total width of the base is 12.5.

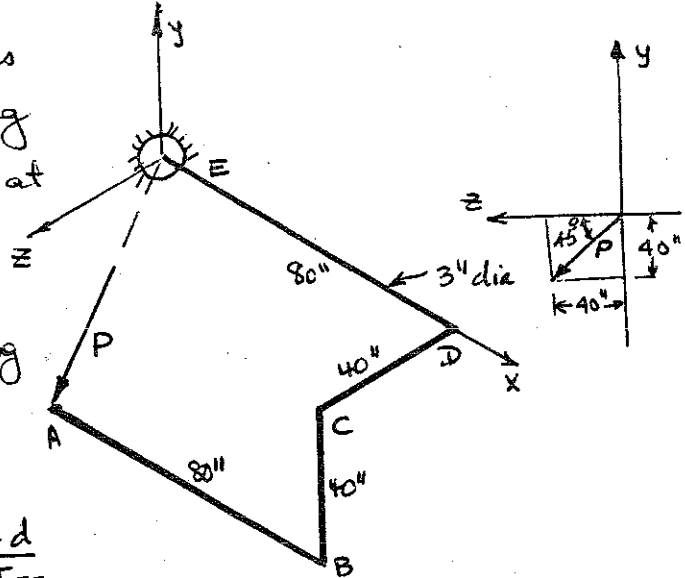
$$\bar{y} = 6.84 \text{ mm} ; A = 44.88 \text{ mm}^2$$

$$\text{Compressive stress} = \frac{300 \text{ N}}{44.88 \times 10^{-6} \text{ m}^2} = 6.68 \text{ MN/m}^2$$

STRESSES too HIGH!

7.62 The rod undergoes combined bending and torsion. If, at a given point z along the rod, the bending and twisting moments are M_b and M_t ,

(Note: $I_x = 2I_{zz}$, where I_x and I_{zz} are about axis & diameter resp.)



7.62 cont

Using the Von Mises yield criterion, yielding will occur when

$$\bar{\sigma} = \sqrt{\frac{1}{2}(\sigma_b^2 + \sigma_t^2) + 3(\tau)^2} = Y = \text{yield strength in tension.}$$

$$\bar{\sigma} = \sqrt{\left(\frac{M_b d}{2I_{zz}}\right)^2 + 3\left(\frac{M_t d}{4I_{zz}}\right)^2} = \frac{M_b d}{2I_{zz}} \left[1 + \frac{3}{4}\left(\frac{M_t}{M_b}\right)^2\right]^{1/2}$$

We will calculate $\bar{\sigma}$ at B, C, and D at one of which it must be maximum; $I_{zz} = I$

- | | |
|--|---|
| 1) just before B: $M_b = 80P, M_t = 0$ | $\bar{\sigma} = \left(\frac{d}{2I}\right) 80P$ |
| 2) just after B: $M_b = \frac{80P}{\sqrt{2}}, M_t = \frac{80P}{\sqrt{2}}$ | $\bar{\sigma} = \left(\frac{d}{2I}\right) 106P$ |
| 3) just before C: $M_b = 63.2P, M_t = 56.8P$ | $\bar{\sigma} = \left(\frac{d}{2I}\right) 87P$ |
| 4) just after C: $M_b = 63.2P, M_t = 56.8P$ | $\bar{\sigma} = \left(\frac{d}{2I}\right) 87P$ |
| 5) just before D: $M_b = \frac{80P}{\sqrt{2}}, M_t = \frac{80P}{\sqrt{2}}$ | $\bar{\sigma} = \left(\frac{d}{2I}\right) 106P$ |
| 6) just after D: $M_b = 80P, M_t = 0$ | $\bar{\sigma} = \left(\frac{d}{2I}\right) 80P$ |

$$(\bar{\sigma})_{\max} = 106P \left(\frac{d}{2I}\right)$$

$$\text{For } 3'' \text{ diameter rod } I = \frac{\pi}{4} \left(\frac{3}{2}\right)^4 = 3.98 \text{ in}^4$$

$$\text{For 4130 HT, } Y = 190,000 \text{ psi}$$

$$\therefore \frac{(106P)(3)}{(2)(3.98)} = 190,000 \quad \text{or} \quad \underline{\underline{P = 4750 \text{ lb}}}$$

(Note: this treatment ignores stress concentrations at corners which may cause the yield load to be reduced considerably)

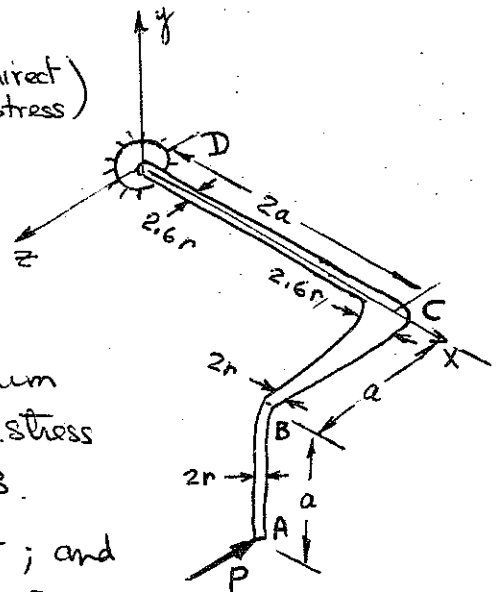
7.63 Stresses are due to combined bending and torsion. (Neglect direct stress)

AB: max. stress at B due to bending.

BC: twisting moment = 0 ;
bending moment = Constant ;
∴ max. stress at minimum section, i.e. at B. This max. stress is the same as that for AB.

CD: twisting moment = Constant ; and
bending moment max. at D.

∴ max. stresses at D.



Yielding when $\bar{\sigma} = Y$ [Von Mises yield criterion] (see prob 7.62). We will use formula for $\bar{\sigma}$ developed in prob 7.62.

$$\bar{\sigma} = \frac{M_b d}{2 I_{yy}} \left[1 + \frac{3}{4} \left(\frac{M_t}{M_b} \right) \right]^{1/2} \quad (\text{where } I = \text{diametral } I)$$

B: $M_b = Pa$; $M_t = 0$; $I = \frac{\pi}{4} r^4$; ∴ $\bar{\sigma} = \sigma_b = \frac{4Pa}{\pi r^3} = 1.26 \frac{Pa}{r^3}$

D: $M_b = 2Pa$; $M_t = Pa$; $I = \frac{\pi}{4} (1.3r)^4$

$$\therefore \bar{\sigma} = \frac{(2Pa)(2.6r)}{2(\frac{\pi}{4})(1.3r)^4} \left[1 + \frac{3}{4} \left(\frac{1}{2} \right) \right]^{1/2} = 1.36 \frac{Pa}{r^3}$$

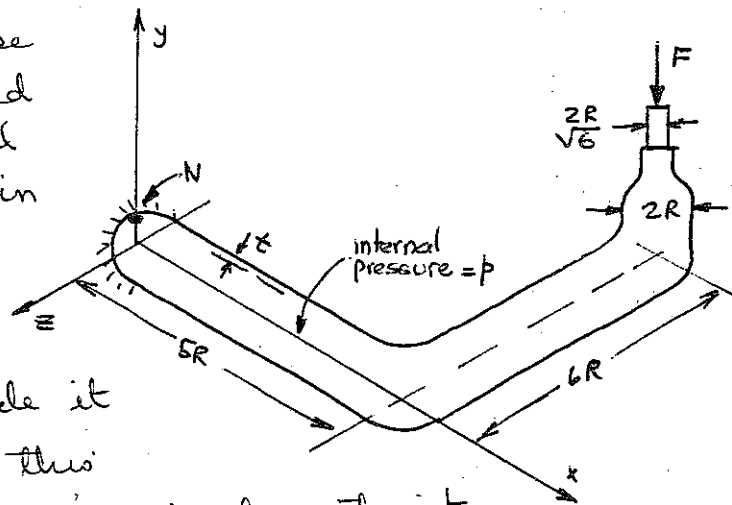
$$\therefore (\bar{\sigma})_{\max} = 1.36 \frac{Pa}{r^3} = Y \text{ for yielding.}$$

$$\therefore \text{Condition for yielding is } P = 0.735 \frac{r^3 Y}{a}$$

If we use the Tresca (max. shear stress) yield criterion, we obtain

$$P = 0.77 \frac{r^3 Y}{a}$$

7.64 GEOMETRY: If the tube is built into a rigid wall, then tangential and radial strains in the tube at the point N (and at similar points) must be zero. While it is possible to solve this



problem under this geometrical constraint, the solution is quite lengthy and involved since the simple theories of bending and internal pressure cannot be used. (these theories require strains in the tangential and radial directions). We will therefore neglect these constraints and use the simple theories of bending etc to evaluate the stresses. In effect, we are calculating the stresses for a point at some distance from the built in end. (It can be shown that this distance is about $\approx R$ when $t/R \approx 1/10$)

Stresses will be found by superimposing those due to bending, torsion, internal pressure, and shear.

EQUILIBRIUM: 1) Piston: $F = p\pi(\frac{R}{\sqrt{6}})^2$ ie $p = \frac{6F}{\pi R^2}$
 2) Tube at built in end: $M_{bz} = -5FR$
 $M_t = -6FR$

GEOMETRY: $I_{yy} = \pi R^3 t$; $I_x = 2\pi R^3 t$

STRESSES: 1) Shear: stress = 0 since $Q = \int y dA = 0$ for N.
 2) Bending: $\sigma_x = \frac{-M_{bz} y}{I_{yy}} = -\frac{(-5FR)R}{\pi R^3 t} = 5F/\pi R t$
 3) Torsion: $\tau_{xz} = M_t r / I_x = (-6FR)(R) / 2\pi R^3 t = -3F/\pi R t$

7.64 CONT

3) Internal pressure :

$$\text{Circumferential stress} = \sigma_z = \frac{pR}{t} = \frac{6F}{\pi R t}$$

$$\text{Axial stress} = \sigma_x = \frac{pR}{2t} = \frac{3F}{\pi R t}$$

TOTAL STRESSES : $\sigma_x = \frac{5F}{\pi R t} + \frac{3F}{\pi R t} = \frac{8F}{\pi R t}$

$$\sigma_z = \frac{6F}{\pi R t}$$

$$\tau_{xz} = -\frac{3F}{\pi R t}$$

(Other stresses are negligible for $(t/R) \ll 1$)

MOHR'S CIRCLE : ($\frac{\text{stress}}{F/\pi R t}$ is plotted) in the $x-z$ plane.

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2}$$

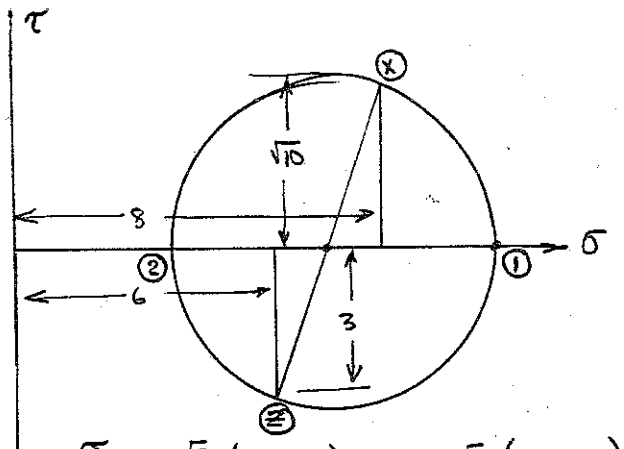
$$\sigma_{\max} = \sigma_1 = \frac{F}{\pi R t} (7 + \sqrt{10})$$

$$\sigma_{\min} = \sigma_y = 0$$

$$\therefore \tau_{\max} = \frac{F}{2\pi R t} (7 + \sqrt{10})$$

For yielding $\tau_{\max} = Y/2$.

$$\therefore F = \frac{\pi R t Y}{7 + \sqrt{10}} = \underline{\underline{0.31 R t Y}}$$



$$\sigma_1 = \frac{F}{\pi R t} (7 + \sqrt{10}) ; \sigma_2 = \frac{F}{\pi R t} (7 - \sqrt{10})$$

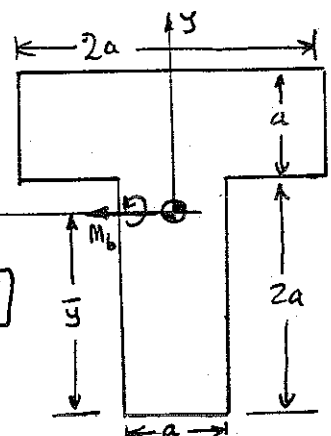
7.65 $M_y = M_b$ when $(\sigma_x)_{\max} = Y$ where σ_x is obtained from the elastic solution.

$$\bar{y} = \frac{(2a)(a)(a) + (2a)(a)(\frac{5a}{2})}{(2a)(a) + (2a)(a)} = \frac{7a^3}{4a^2} = \frac{7a}{4}$$

$$I_{zz} = \left[\frac{1}{12}(a)(2a)^3 + (2a)(a)\left(\frac{3a}{4}\right)^2 \right] + \left[\frac{1}{12}(2a)(a)^3 + (2a)(a)\left(\frac{3a}{4}\right)^2 \right]$$

$$= \frac{37}{12} a^4$$

$$\text{And } y_{\max} = \frac{7}{4} a$$



1.65 cont

$$(\sigma_x)_{\max} = \frac{(M_b) \left(\frac{7a}{4} \right)}{\frac{37}{12} a^4} = Y \quad \therefore M_b = M_Y = \left(\frac{37}{12} \right) \left(\frac{4}{7} \right) Y a^3 = \frac{37}{21} Y a^3$$

Fully Plastic: $\sigma_x = \pm Y$ except at the neutral axes where $\sigma_x = 0$.

EQUILIBRIUM

$$\sum F_x = 0 : -Y A_1 + Y A_2 = 0$$

$$\sum M_z = 0 : M_b + \int \sigma_x y dA = 0$$

The first equation locates the neutral axis, giving $y_N = 2a$.

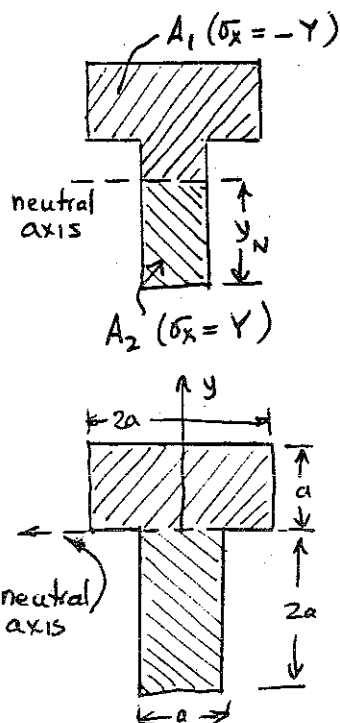
The second equation gives $M_b (= M_L)$.

For an area A_i under uniform stress σ , with centroid located at y_i

$$\int \sigma_x y dA = \sigma A_i y_i$$

$$\begin{aligned} \text{Thus: } M_b &= - \int \sigma_x y dA = -(-Y)(2a^2)\left(\frac{a}{2}\right) \\ &\quad - (Y)(2a^2)(-a) \\ &= 3Ya^3 = M_L \end{aligned}$$

$$\therefore K = M_L / M_Y = \frac{3Ya^3}{\frac{37}{21} Ya^3} = \underline{\underline{1.70}}$$



7.66 Consider half the section, to find half the limit moment.

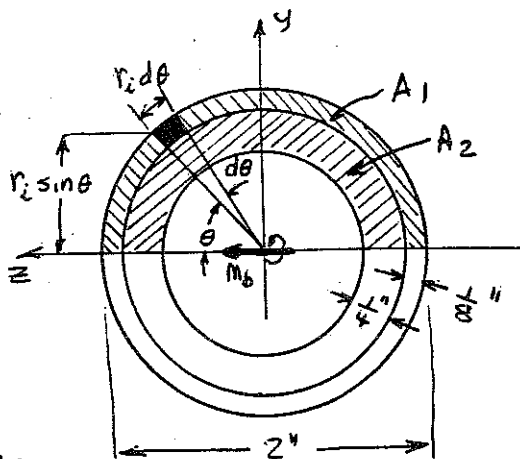
Each of the areas A_1 and A_2 are under uniform stress equal to $-Y_1$ and Y_2 respectively.

$$\sum M_z = 0 : \frac{M_b}{2} = \frac{M_L}{2} = - \int_A \sigma_x y dA =$$

$$Y_1 \int_{A_1} y dA + Y_2 \int_{A_2} y dA.$$

Approximate calculation of $\int y dA$:

$$\text{Area ①: } \int y dA = \int_0^\pi (r_1 d\theta)(\bar{r}_1)(r_1 \sin\theta)$$



7-66 cont

where $r_1 = \text{mean radius for } \textcircled{1} = 1 - \frac{1}{16} = \frac{15}{16}''$
 $t_1 = \text{thickness for } \textcircled{1} = \frac{1}{8}''$

$$\int_{A_1} y dA = \left(\frac{15}{16}\right)^2 \left(\frac{1}{8}\right) \int_0^\pi \sin \theta d\theta = 2 \left(\frac{15}{16}\right)^2 \left(\frac{1}{8}\right) = \underline{0.22 \text{ in}^3}$$

$$\text{And } \int_{A_2} y dA = \left(\frac{12}{16}\right)^2 \left(\frac{1}{4}\right) \int_0^\pi \sin \theta d\theta = 2 \left(\frac{12}{16}\right)^2 \left(\frac{1}{4}\right) = \underline{0.28 \text{ in}^3}$$

and $Y_1 = 37000 \text{ psi}$ $Y_2 = 14000 \text{ psi}$

$$M_L = 2 \left[(37000)(0.22) + (14000)(0.28) \right] = \underline{\underline{24,150 \text{ in-lb}}}$$

7.67 For 2024-T4 Aluminum, $Y = 55,000 \text{ psi}$
 and $\epsilon_Y = \frac{55000}{11 \times 10^6} = 0.005 = \text{yield strain.}$

The ϵ_x strains are directly proportional to the distance from the neutral axis. Therefore if the strain in $\textcircled{4}$ is 0.016 , that in $\textcircled{1}$ is -0.016 , that in $\textcircled{2}$ is $-0.016\left(\frac{1}{2}\right) = -0.008$ and that in $\textcircled{3}$ is 0.008 . These are

all greater than ϵ_Y and therefore all the strips are in the plastic state. (The moment M_b is the limit moment M_L .)

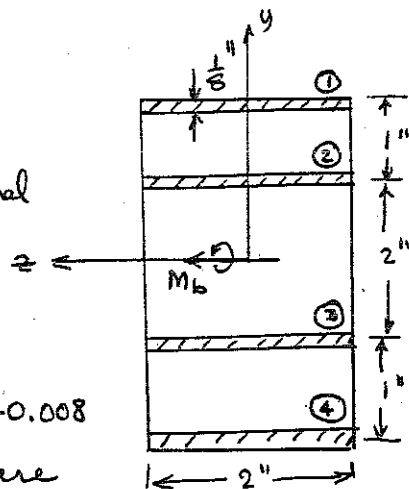
STRESSES: $(\sigma_x)_1 = -55000 \text{ psi}$; $(\sigma_x)_2 = -55000 \text{ psi}$; $(\sigma_x)_3 = 55000 \text{ psi} = (\sigma_x)_4$

$$\sum M_z = 0: M_b = -\int \sigma_x y dA = -\sum_i (\sigma_x)_i A_i \bar{y}_i \quad (\text{see prob 7.66})$$

$$= (2)(55000) \left[2\left(\frac{1}{8}\right)(2) + 2\left(\frac{1}{8}\right)(1) \right] = \underline{\underline{82,500 \text{ in-lb.}}}$$

UNLOADING: We assume that the unloading is elastic (to be checked later) and add the elastic stresses due to moment $-M_b$ to the loading stresses (above) to get the residual stresses.

$$I_{zz} = 2 \left[(2)\left(\frac{1}{8}\right)(2)^2 + (2)\left(\frac{1}{8}\right)(1)^2 \right] = 2.5 \text{ in}^4$$



7.67 cont

Elastic stress due to moment $-M_b$

$$(\sigma_x)_i = - \frac{(-M_b)y_i}{I_{zz}} = \frac{M_b y_i}{I_{zz}}$$

$$\text{Thus } (\sigma_x)_1 = \frac{(82,500)(2)}{2.5} = 66,000 \text{ psi}$$

$$(\sigma_x)_4 = -66,000 \text{ psi}$$

$$(\sigma_x)_2 = \frac{(82,500)(1)}{2.5} = 33,000 \text{ psi}$$

$$(\sigma_x)_3 = -33,000 \text{ psi}$$

RESIDUAL STRESSES $(\sigma_x)_1 = -55,000 + 66,000 = 11,000 \text{ psi}$

$$(\sigma_x)_2 = -55,000 + 33,000 = -22,000 \text{ psi}$$

$$(\sigma_x)_3 = 55,000 - 33,000 = 22,000 \text{ psi}$$

$$(\sigma_x)_4 = 55,000 - 66,000 = -11,000 \text{ psi}$$

(Since these residual stresses are all below Y , our elastic unloading assumption is justified).

7.68 Wire diameter $= d = 0.036''$

From postage scale test : $M_L = 0.61 \text{ in-lb}$

Elastic solution: for initial yielding, $M_b = M_Y = \left(\frac{\pi d^4}{64}\right)\left(\frac{2}{d}\right)Y = \frac{\pi d^3 Y}{32}$

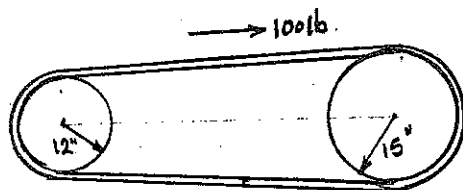
From table 7.1 : $\frac{M_L}{M_Y} = 1.3$ for solid circle.

$$\therefore M_L = 1.3 M_Y = \frac{(1.3)\pi d^3 Y}{32} = 0.61 \text{ in-lb}$$

$$\text{ie } Y = \frac{(0.61)(32)}{(1.3)(\pi)(0.036)^3} = 102,000 \text{ psi}$$

The material is probably 1020CR steel in a work hardened state.

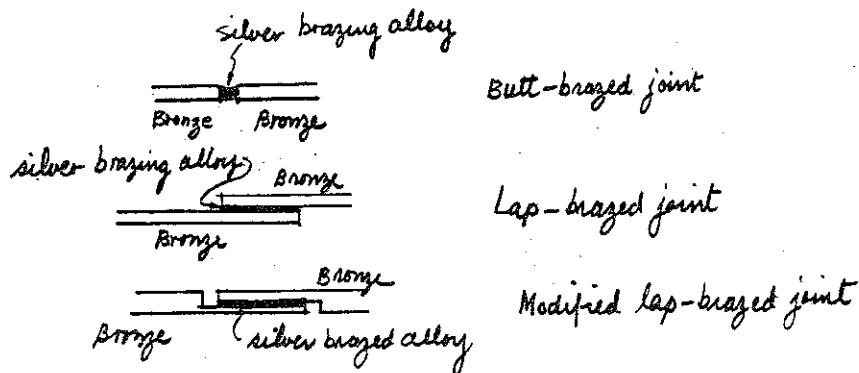
(7-69)



FROM PROBLEM #7.15

$$\epsilon_x = \frac{y}{\rho}$$

$$(\epsilon_x)_{\max} = \frac{h}{2\rho} = \frac{1}{2 \times 300} = 1.667 \times 10^{-3}$$



Butt-brazed joint

Lap-brazed joint

Modified lap-brazed joint

In the brazing alloy (neglect tensile stress)

$$(\sigma_x)_{\max} = (\epsilon_x)_{\max} E_{\text{Alloy}} = 1.67 \times 10^{-3} \times 78 \times 10^3 = 130 \text{ MN/m}^2 > 105 \text{ MN/m}^2$$

IN THE PHOSPHOR BRONZE

$$(\sigma_x)_{\max} = (\epsilon_x)_{\max} E_{\text{P.B.}} = 1.67 \times 10^{-3} \times 110 \times 10^3 = 183 \text{ MN/m}^2 < 220$$

∴ THERE IS ORIGINALLY FATIGUE IN BRAZING ALLOY

$$(a) \quad h = 1.2 \text{ mm} \quad \epsilon_x = \frac{1.2}{600} = 2 \times 10^{-3}$$

$$\text{Alloy: } (\sigma_x)_{\max} = 156 \text{ MN/m}^2 > 105 \text{ MN/m}^2$$

$$\text{P.B. : } (\sigma_x)_{\max} = 220 \text{ MN/m}^2 = 220 \text{ MN/m}^2 \quad ; \text{ No good}$$

$$(b) \quad h = 0.3 \quad \epsilon_x = \frac{0.3}{600} = 0.5 \times 10^{-3} \quad ; \text{ TENSILE STRESS} = \frac{F}{0.3 \times 25} = \frac{F}{7.5}$$

$$\text{Alloy: } (\sigma_x)_{\max} = 39 + 60 = \text{BENDING} + \text{TENSILE} = 99 < 105 \text{ MN/m}^2$$

$$\text{P.B. : } (\sigma_x)_{\max} = 55 + 60 = 115 < 220 \text{ MN/m}^2$$

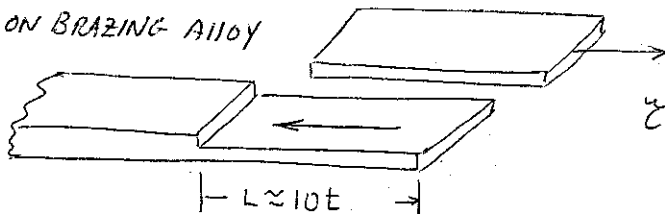
WON'T FATIGUE, DESIGN OK - JUST!

7-69 (Cont'd)

(c) The lap-brazed joint is equivalent to doubling the thickness of the belt at the joint, $\therefore (\epsilon_x)_{\max}$ is same as in (a) and phosphor bronze will fatigue.

(d) $(\epsilon_x)_{\max}$ from bending same as in original case. Phosphor bronze will not fatigue.

CHECK SHEAR ON BRAZING ALLOY



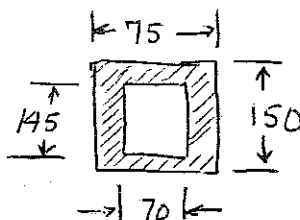
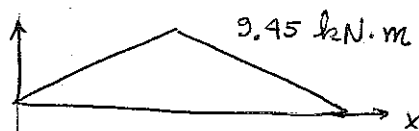
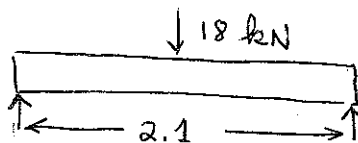
$$\tau = \frac{F}{A} = \frac{450 \text{ N}}{25 \times 10 \times 10^{-6} \text{ m}^2} = 1.8 \text{ MN/m}^2$$

$\therefore$ DESIGN WON'T FATIGUE, OK

(e) SINCE BAND THICKNESS AND JOINT TYPE SAME AS ORIGINAL DESIGN, IT WILL FAIL THE SAME WAY.

DESIGN (b) + (d) ARE O.K WITH (b) EASIER TO MANUFACTURED

7-70



$$t = 2.5 \text{ mm}$$

$$I = \frac{b_1 h_1^3}{12} - \frac{b_2 h_2^3}{12} = 3.31 \times 10^6 \text{ mm}^4$$

For (b) $\sigma_x = -\frac{My}{I} = -\frac{9.45 \times 10^3 \times 75 \times 10^{-3}}{3.31 \times 10^{-6}} = -214 \text{ MN/m}^2$

For (c) $V = 9 \text{ kN}$

$$b = 2.5 \text{ mm}$$

$$I = 3.31 \times 10^{-6} \text{ m}^4$$

$$Q = 1.43 \times 10^{-5} \text{ m}^3$$

$$\tau_{xy} = -\frac{VQ}{bI} = \frac{9 \times 1.43 \times 10^{-5}}{2.5 \times 10^{-3} \times 3.31 \times 10^{-6}} = 15.5 \text{ MN/m}^2$$

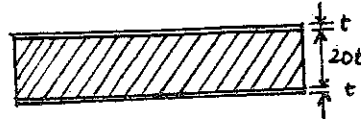
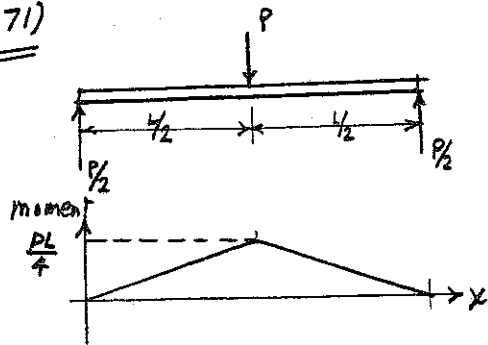
For (d) $Q = 75 \times 75 \times 37.5 - (70 \times 72.5 \times 36.25)$
 $= 2.7 \times 10^{-5} \text{ m}^3$

$$\tau_{xy} = 29.3 \text{ MN/m}^2$$

(b) IS THE BEST DESIGN

7-50

(7-71)



$$E_w = 0.06 E_m$$

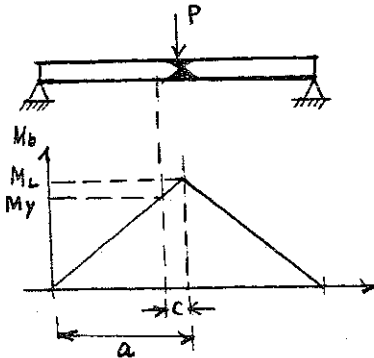
$$(I_{zz})_w = \frac{bh^3}{12} = \frac{150 \times (20t)^3}{12} = 1 \times 10^5 t^3$$

$$(I_{zz})_m = 2 \left(\frac{bh^3}{12} + Ad^2 \right) = 2 \left[\frac{150t^3}{12} + 150t \left(\frac{21t}{2} \right)^2 \right] = 3175 t^3$$

$$(\sigma_x)_m = - \frac{E_m M_b}{E_m (I_{zz})_m + E_w (I_{zz})_w} = - \frac{PL}{3.67 \times 10^4 t^3}$$

$$(\sigma_x)_w = - \frac{E_w M_b}{E_m (I_{zz})_m + E_w (I_{zz})_w} = - \frac{PL}{6.12 \times 10^5 t^3}$$

(7-72)



$$\frac{a-c}{a} = \frac{M_y}{M_L} = \frac{M_y}{\frac{3}{2} M_y}$$

$$\frac{c}{a} = 1 - \frac{2}{3} = \frac{1}{3}$$

7-73

From Example 7.9:

$$\frac{d\alpha}{ds_1} = 1.36 \frac{M_{bz}}{Ec^4}$$

$$\frac{d\beta}{ds_2} = -1.58 \frac{M_{bz}}{Ec^4}$$

$$\text{Eqn. 7.53: } \frac{d\alpha}{ds_1} = \frac{d\phi}{ds} \cos \theta$$

$$\frac{d\beta}{ds_2} = \frac{d\phi}{ds} \sin \theta$$

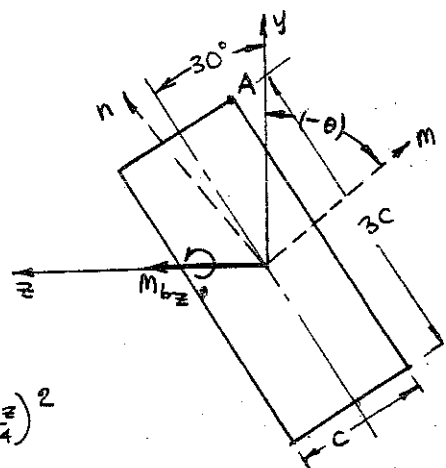
$$\therefore \left(\frac{d\phi}{ds} \right)^2 = \left(\frac{d\alpha}{ds_1} \right)^2 + \left(\frac{d\beta}{ds_2} \right)^2 = 4.35 \left(\frac{M_{bz}}{Ec^4} \right)^2$$

$$\therefore \frac{d\phi}{ds} = 2.09 \frac{M_{bz}}{Ec^4} ; \tan \theta = \frac{d\beta/ds_2}{d\alpha/ds_1} = -1.16$$

giving $\theta = -49.2^\circ$; Coordinates of A: $y = 1.55c$; $z = 0.32c$ Transformation to m, n coordinates: $m = 0.775c$; $n = 1.38c$

$$(\sigma_x)_A = -E m \left(\frac{d\phi}{ds} \right) = -E (0.775c) (2.09 \frac{M_{bz}}{Ec^4}) = -1.62 \frac{M_{bz}}{c^3}$$

7-51

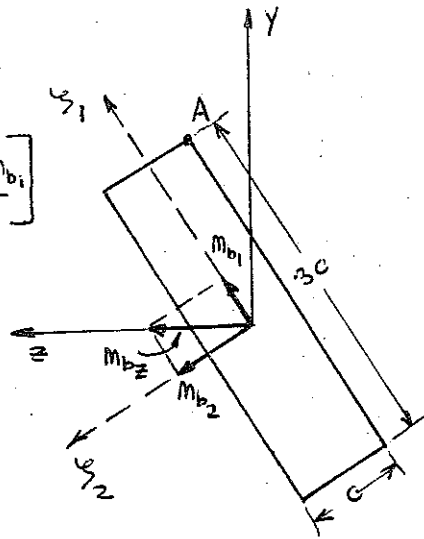


For arbitrary axes ζ_1 and ζ_2 ,
Eqn 7.14 gives

$$\sigma_x = - \left[\frac{(\zeta_1 I_{22} - \zeta_2 I_{12}) M_{b2} + (\zeta_1 I_{12} - \zeta_2 I_{11}) M_{b1}}{I_{11} I_{22} - I_{12}^2} \right]$$

Now if ζ_1 and ζ_2 are principal axes, then $I_{12} = 0$, and this reduces to:

$$\begin{aligned} \sigma_x &= - \left[\frac{\zeta_1 M_{b2}}{I_{11}} - \frac{\zeta_2 M_{b1}}{I_{22}} \right] \\ &= \frac{M_{b1}}{I_{22}} \zeta_2 - \frac{M_{b2}}{I_{11}} \zeta_1 \end{aligned}$$



Coordinates of point A: $\zeta_1 = 1.5c$ $\zeta_2 = -0.5c$

$$I_{22} = \frac{1}{12} (c)(3c)^3 = \frac{9}{4} c^4$$

$$I_{11} = \frac{1}{12} (3c)(c)^3 = \frac{1}{4} c^4$$

$$M_{b1} = M_{bz} \cos 60^\circ = 0.5 M_{bz}$$

$$M_{b2} = M_{bz} \sin 60^\circ = 0.866 M_{bz}$$

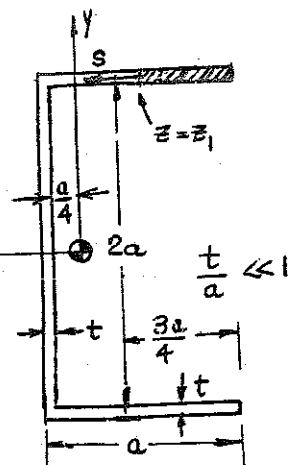
$$\therefore \sigma_x = \frac{0.5 M_{bz}}{\frac{1}{4} c^4} (-0.5c) - \frac{0.866 M_{bz}}{\frac{9}{4} c^4} (1.5c) = -1.60 \frac{M_{bz}}{c^3}$$

Symmetry: Shear center must lie on z axis, and $I_{yz} = 0$

We will first determine the shear flow $q_{xz} = q_{xs}$ due to a y -direction force V . The resultant of these shear flows must pass through the shear center.

First consider the shaded area (shown detached) in the upper horizontal leg.

$$Q = \text{first moment of area} = \left(z_1 + \frac{3a}{4} \right) (t)(a)$$



$$I = I_{zz}, \text{ throughout}$$

(7-75 cont)

$$\therefore q_{xs} = -\frac{VQ}{I} = -\frac{V}{I} \text{ at } (z_1 + \frac{3a}{4}) \quad \left\{ -\frac{3a}{4} \leq z_1 \leq \frac{a}{4} ; y = a \right\}$$

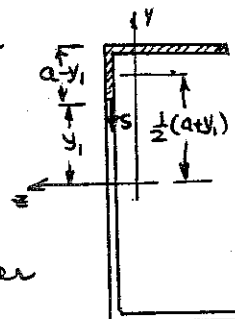
Similarly, for the corresponding area on the lower horizontal leg,

$$q_{xs} = -\frac{V}{I} \text{ at } (z_1 + \frac{3a}{4}) \quad \left\{ -\frac{3a}{4} \leq z_1 \leq \frac{a}{4} ; y = -a \right\}$$

(Note that the positive 's' direction here is the $-z$ direction, while it is the positive z direction for the upper leg)

Next, for the area shown in the figure alongside,

$$Q = (a^2 t) + (a - y_1)(t) \left(\frac{a + y_1}{2} \right) = \frac{3a^2 t}{2} - \frac{t}{2} y_1^2 ; q_{xs} = -\frac{VQ}{I}$$



Resultant z direction force in the upper horizontal leg:

$$F_{1z} = \int_{z_1 = -\frac{3a}{4}}^{\frac{a}{4}} q_{xs} dz = \int_{-\frac{3a}{4}}^{\frac{a}{4}} -\frac{V}{I} \text{ at } (z + \frac{3a}{4}) dz = -\frac{V}{I} \left(\frac{a^3 t}{2} \right)$$

Similarly, for the lower horizontal leg,

$$F_{2z} = \int_{-\frac{3a}{4}}^{\frac{a}{4}} (-q_{xs}) dz = + \int_{-\frac{3a}{4}}^{\frac{a}{4}} \frac{V}{I} \text{ at } (z + \frac{3a}{4}) dz = +\frac{V}{I} \left(\frac{a^3 t}{2} \right)$$

Resultant y -direction force for the vertical leg =

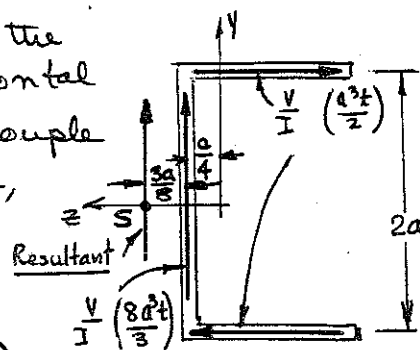
$$F_{3y} = \int_{-a}^a (-q_{xs}) dy = +\frac{V}{I} \int_{-a}^a \left(\frac{3a^2 t}{2} - \frac{t}{2} y^2 \right) dy = \frac{V}{I} \left(\frac{8a^3 t}{3} \right)$$

These resultants are shown in the figure alongside. The two horizontal forces result in a clockwise couple

$$= \frac{V}{I} \left(\frac{a^3 t}{2} \right) (2a) = \frac{V}{I} (a^4 t). \text{ This couple,}$$

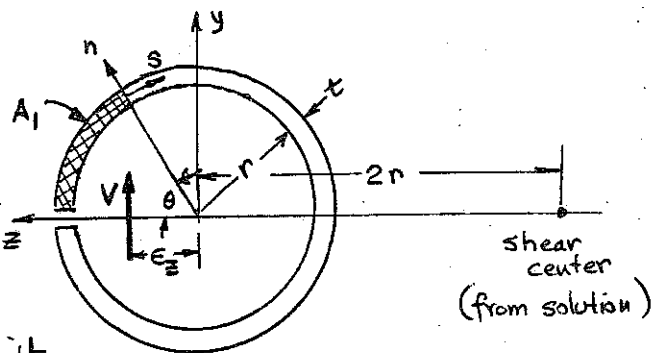
and the vertical force result in a vertical force at distance "d" to the left of the vertical leg

$$\text{where } d = \frac{V}{I} (a^4 t) / \frac{V}{I} \left(\frac{8a^3 t}{3} \right) = (3a/8).$$



76

Symmetry: Shear center lies on the z axis; $I_{yz}=0$.
Let the shear center be located at a distance ϵ_z to the left of the origin, with a vertical force V passing through it.



Equilibrium: $\sum M_x = 0$: Moment of V about the origin = moment of shear flow q_{xs} about origin.

For hatched area A_1 : $Q = \int y dA = \int_{\phi=0}^{\phi=\theta} (r \sin \phi)(t r d\phi) = r^2 t (1 - \cos \theta)$

$\therefore q_{xs} = \frac{-V}{I_{zz}} r^2 t (1 - \cos \theta)$

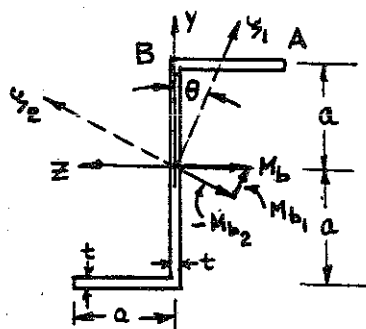
Moment of q_{xs} about origin = $-\int_{\theta=0}^{\theta=2\pi} q_{xs} (r d\theta) (r)$
 $= + \int_0^{2\pi} \frac{V}{I_{zz}} r^4 t (1 - \cos \theta) d\theta = \frac{V}{I_{zz}} (2\pi r^4 t)$

Moment of $V = -V \epsilon_z$

$\therefore -V \epsilon_z = \frac{V}{I_{zz}} (2\pi r^4 t) \quad \text{ie } \epsilon_z = - \frac{2\pi r^4 t}{I}$

$I_{zz} = \pi r^3 t$ (see prob 7.16); $\therefore \epsilon_z = - \frac{2\pi r^4 t}{\pi r^3 t} = \underline{\underline{-2r}}$

77

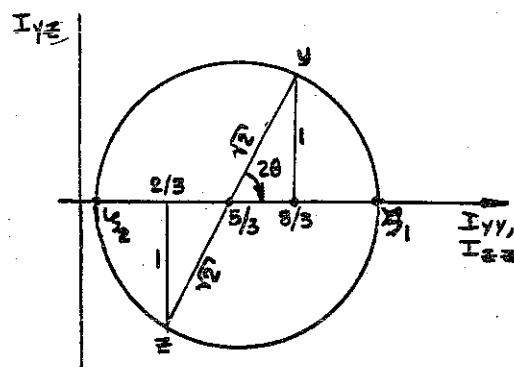


$I_{zz} = \frac{8}{3} a^3 t$

$I_{yy} = \frac{2}{3} a^3 t$

$I_{yz} = -a^3 t$

Mohr's circle: ($I/a^3 t$ is plotted)



From Mohr's circle:

I_{z2} (corresponding to y_1) = $3.08 a^3 t$

$I_{11} = 0.26 a^3 t$; $\theta = 22.5^\circ$

7-54

7.77 cont

We will use the result of prob. 7.49 :-

$$\sigma_x = \frac{M_{b1}}{I_{11}} \eta_2 - \frac{M_{b2}}{I_{22}} \eta_1$$

Here $M_{b1} = M_b \sin 22.5^\circ = 0.384 M_b$

$M_{b2} = -M_b \cos 22.5^\circ = -0.923 M_b$

Max. stress will occur either at A or at B.

Coordinates:

A : $\eta_1 = 1.307a$ B : $\eta_1 = 0.923a$

$\eta_2 = -0.539a$ $\eta_2 = 0.384a$

$$(\sigma_x)_A = \frac{(0.384 M_b)(-0.539a)}{(0.26 ta^3)} - \frac{(-0.923 M_b)(1.307a)}{(3.08 ta^3)} = -0.406 \frac{M_b}{ta^2}$$

$$(\sigma_x)_B = \frac{(0.384 M_b)(0.384a)}{(0.26 ta^3)} - \frac{(-0.923 M_b)(0.923a)}{(3.08 ta^3)} = 0.844 \frac{M_b}{ta^2}$$

$\therefore (\sigma_x)_{\max} = 0.844 \frac{(M_b)_{\max}}{ta^2}$; and $(M_b)_{\max}$ can be easily

shown to be PL

$$\therefore (\sigma_x)_{\max} = \frac{0.844 PL}{ta^2}$$

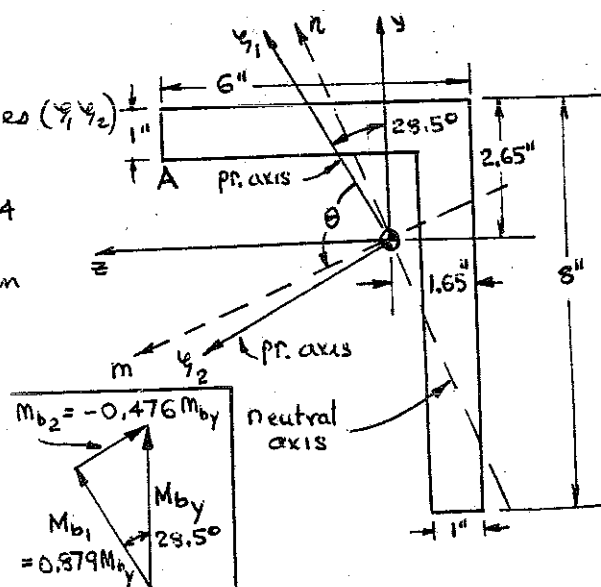
7.78

From probs 7.2, 7.10:

Centroid and principal axes (η_1, η_2) located as shown, and

$$I_{22} = 98.2 \text{ in}^4 ; I_{11} = 21.4 \text{ in}^4$$

One may solve the problem either by using a formula for σ_x (such as that developed in prob 7.49) and testing all 6 corners, or by locating the neutral axis "n" and finding (on a scale drawing) the point



farthest from it. This point will experience max. σ_x which can be determined by an appropriate formula.

Egns 7.⁶³ (text): change 1 to 1 and 2 to 2 and note that $I_{12} = 0$ since 1, 2 are principal axes.

giving: $\frac{d\phi}{ds} \cos\theta = \frac{M_{b2}}{EI_{22}}$ where θ is the counterclockwise angle from the 1 axis to the m axis

$\frac{d\phi}{ds} \sin\theta = -\frac{M_{b1}}{EI_{11}}$

Thus $\tan\theta = -\frac{M_{b1}I_{22}}{M_{b2}I_{11}} = -\frac{(0.879M_{by})(98.2)}{(-0.476M_{by})(21.4)} = 8.5$

$\theta = 83.3^\circ$ locating the m, n axes as shown.

The point with the max. m coordinate is A.

For A: $\psi_1 = 3.53"$
 $\psi_2 = 3.03"$

and $(\sigma_x)_A = (\sigma_x)_{\max} = \frac{M_{b1}\psi_2}{I_{11}} - \frac{M_{b2}\psi_1}{I_{22}} = \frac{(0.879M_{by})(3.03)}{21.4} - \frac{(-0.476M_{by})(3.53)}{98.2}$
 $= 0.141M_{by} = 0.141(100,000) = \underline{\underline{14,100 \text{ psi}}}$

Symmetry: shear center lies on the z axis and $I_{yz} = 0$.

$$q_{xs} = q_{sx} = \frac{-V_y}{I_{zz}} \int y dA$$

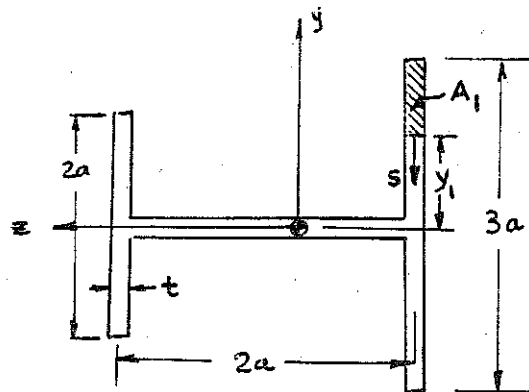
(for a vertical force V_y)

The horizontal section has

$q_{xs} = 0$ everywhere since

$\int y dA \approx 0$ at all points.

Right hand vertical leg: $\int_{A_1} y dA = \int_{y_1}^{3a/2} y(t dy) = \frac{t}{2} \left(\frac{9a^2}{4} - y_1^2 \right)$



7-79 cont

and $q_{xs} = -\frac{V_y t}{2I} \left(\frac{qa^2}{4} - y_1^2 \right)$ $I = I_{zz}$

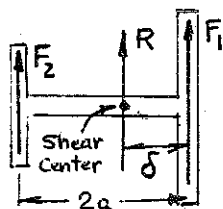
The total shear force on this leg is $\int_{-3a/2}^{3a/2} (-q_{xs}) dy = F_1$
(positive in y direction)

$$= \frac{V_y t}{2I} \int_{-3a/2}^{3a/2} \left(\frac{qa^2}{4} - y^2 \right) dy = \frac{V_y t}{2I} \left(\frac{9}{2} a^3 \right) = F_1$$

Left hand vertical leg: $\int y dA = \frac{t}{2} (a^2 - y_1^2)$

and $F_2 = \frac{V_y t}{2I} \int_{-a}^a (a^2 - y^2) dy = \frac{V_y t}{2I} \left(\frac{4}{3} a^3 \right)$

The resultant of F_1 and F_2 passes through the shear center.



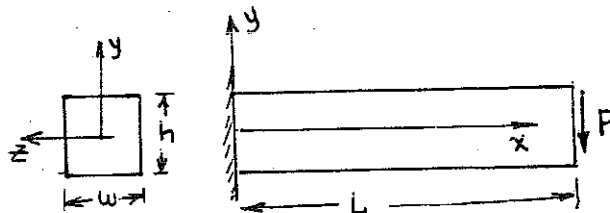
$$\sum M_{\text{shear center}} = 0$$

$$\frac{V_y t}{2I} \left(\frac{9}{2} a^3 \right) (\delta) - \frac{V_y t}{2I} \left(\frac{4}{3} a^3 \right) (2a - \delta) = 0$$

giving $\delta = \frac{16}{35} a$

7-80

When (w/h) is small, and the load P is applied in the proper manner, we will have plane stress in the x - y plane.



at $y = h/2$: $\sigma_x = \text{tension}$; $\sigma_y = 0$; $\tau_{xy} = 0$

∴ x is a principal tension direction

y is a principal compression direction.

at $y = -h/2$: $\sigma_x = \text{compression}$; $\sigma_y = 0$; $\tau_{xy} = 0$

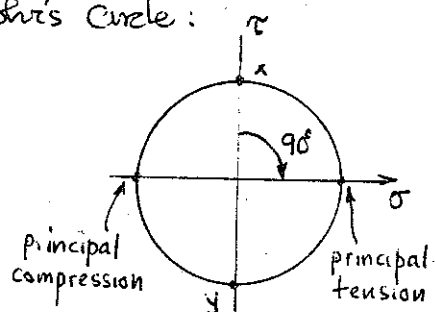
∴ x is a principal compression direction

z is a principal tension direction.

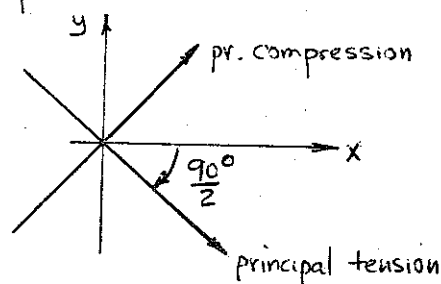
80 CONT

at $y=0$: $\sigma_x=0$; $\sigma_y=0$; τ_{xy} = negative

Mohr's Circle:

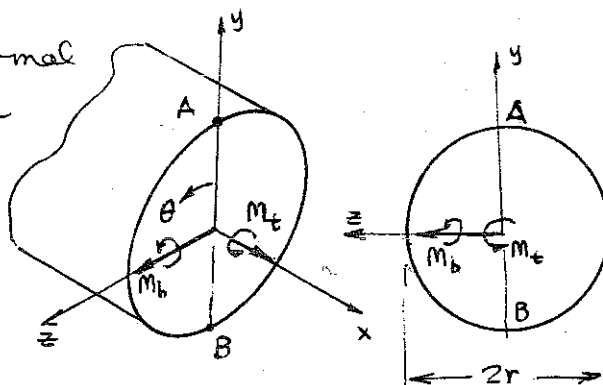


Principal directions -



These illustrations are sufficient to demonstrate the validity of the trajectories shown.

Bending causes a normal stress σ_x , and torsion results in a shear stress τ_{xz} , where θ is the tangential direction. These stresses are max.



at A and B and $\therefore$ max. shear stress will occur at these points.

Consider stresses at A :

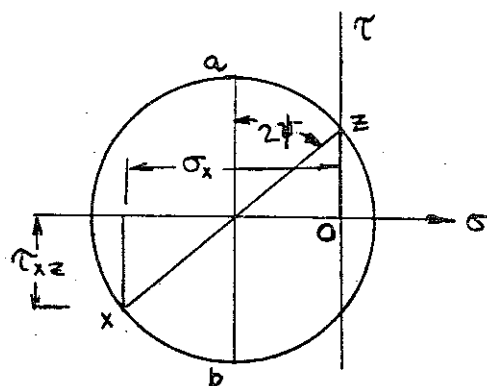
$$\sigma_x = - \frac{(M_b)(r)}{I_{zz}}$$

Mohr's circle in the x-z plane

$$\tau_{xz} = \tau_{x,z} = \frac{(M_t)(r)}{I_x} = \frac{M_t r}{2I_{zz}}$$

$$\left(\text{since } I_x = \frac{\pi r^4}{2} = 2 \left(\frac{\pi r^4}{4} \right) = 2I_{zz} \right)$$

Other stresses are zero.



7-81 CONT

Points "a" and "b" on the Mohr's circle represent directions that are normal to the planes of max. shear stress. Corresponding planes are normal to a, b.

$$\tau_{\max} = \left[\left(\frac{\sigma_x}{2} \right)^2 + (\tau_{xz})^2 \right]^{\frac{1}{2}} = \left[\left(\frac{-M_b r}{2I_{zz}} \right)^2 + \left(\frac{M_t r}{2I_{zz}} \right)^2 \right]^{\frac{1}{2}} = \frac{r}{2I} \sqrt{(M_b)^2 + (M_t)^2}$$

$$\tan 2\psi = \frac{-\sigma_x/2}{\tau_{xz}} = \frac{M_b r}{2I_{zz}} \frac{2I_{yz}}{M_t r} = \frac{M_b}{M_t}$$

$$\text{or } \psi = \frac{1}{2} \tan^{-1} \left(\frac{M_b}{M_t} \right)$$

Hollow shaft: Max. stress occurs

at the same points as for the solid shaft and $I_x = 2I_{zz}$ (by symmetry and prob. 7.4) which

is all that was required to prove the above result. It is

therefore valid for the hollow shaft also.

